

Appendix

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I Cumulative Standard Normal Distribution^a

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$$

<i>z</i>	0.00	0.01	0.02	0.03	0.04	<i>z</i>
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.0
0.1	0.53983	0.54379	0.54776	0.55172	0.55567	0.1
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.2
0.3	0.61791	0.62172	0.62551	0.62930	0.63307	0.3
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.4
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.5
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.6
0.7	0.75803	0.76115	0.76424	0.76730	0.77035	0.7
0.8	0.78814	0.79103	0.79389	0.79673	0.79954	0.8
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.9
1.0	0.84134	0.84375	0.84613	0.84849	0.85083	1.0
1.1	0.86433	0.86650	0.86864	0.87076	0.87285	1.1
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	1.2
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	1.3
1.4	0.91924	0.92073	0.92219	0.92364	0.92506	1.4
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	1.5
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	1.6
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	1.7
1.8	0.96407	0.96485	0.96562	0.96637	0.96711	1.8
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	1.9
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	2.0
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	2.1
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	2.2
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	2.3
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	2.4
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	2.5
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	2.6
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	2.7
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	2.8
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	2.9
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	3.0
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	3.1
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	3.2
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	3.3
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	3.4
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	3.5
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	3.6
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	3.7
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	3.8
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	3.9

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I Cumulative Standard Normal Distribution (*Continued*)

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$$

<i>z</i>	0.05	0.06	0.07	0.08	0.09	<i>z</i>
0.0	0.51994	0.52392	0.52790	0.53188	0.53586	0.0
0.1	0.55962	0.56356	0.56749	0.57142	0.57534	0.1
0.2	0.59871	0.60257	0.60642	0.61026	0.61409	0.2
0.3	0.63683	0.64058	0.64431	0.64803	0.65173	0.3
0.4	0.67364	0.67724	0.68082	0.68438	0.68793	0.4
0.5	0.70884	0.71226	0.71566	0.71904	0.72240	0.5
0.6	0.74215	0.74537	0.74857	0.75175	0.75490	0.6
0.7	0.77337	0.77637	0.77935	0.78230	0.78523	0.7
0.8	0.80234	0.80510	0.80785	0.81057	0.81327	0.8
0.9	0.82894	0.83147	0.83397	0.83646	0.83891	0.9
1.0	0.85314	0.85543	0.85769	0.85993	0.86214	1.0
1.1	0.87493	0.87697	0.87900	0.88100	0.88297	1.1
1.2	0.89435	0.89616	0.89796	0.89973	0.90147	1.2
1.3	0.91149	0.91308	0.91465	0.91621	0.91773	1.3
1.4	0.92647	0.92785	0.92922	0.93056	0.93189	1.4
1.5	0.93943	0.94062	0.94179	0.94295	0.94408	1.5
1.6	0.95053	0.95154	0.95254	0.95352	0.95448	1.6
1.7	0.95994	0.96080	0.96164	0.96246	0.96327	1.7
1.8	0.96784	0.96856	0.96926	0.96995	0.97062	1.8
1.9	0.97441	0.97500	0.97558	0.97615	0.97670	1.9
2.0	0.97982	0.98030	0.98077	0.98124	0.98169	2.0
2.1	0.98422	0.98461	0.98500	0.98537	0.98574	2.1
2.2	0.98778	0.98809	0.98840	0.98870	0.98899	2.2
2.3	0.99061	0.99086	0.99111	0.99134	0.99158	2.3
2.4	0.99286	0.99305	0.99324	0.99343	0.99361	2.4
2.5	0.99461	0.99477	0.99492	0.99506	0.99520	2.5
2.6	0.99598	0.99609	0.99621	0.99632	0.99643	2.6
2.7	0.99702	0.99711	0.99720	0.99728	0.99736	2.7
2.8	0.99781	0.99788	0.99795	0.99801	0.99807	2.8
2.9	0.99841	0.99846	0.99851	0.99856	0.99861	2.9
3.0	0.99886	0.99889	0.99893	0.99897	0.99900	3.0
3.1	0.99918	0.99921	0.99924	0.99926	0.99929	3.1
3.2	0.99942	0.99944	0.99946	0.99948	0.99950	3.2
3.3	0.99960	0.99961	0.99962	0.99964	0.99965	3.3
3.4	0.99972	0.99973	0.99974	0.99975	0.99976	3.4
3.5	0.99981	0.99981	0.99982	0.99983	0.99983	3.5
3.6	0.99987	0.99987	0.99988	0.99988	0.99989	3.6
3.7	0.99991	0.99992	0.99992	0.99992	0.99992	3.7
3.8	0.99994	0.99994	0.99995	0.99995	0.99995	3.8
3.9	0.99996	0.99996	0.99996	0.99997	0.99997	3.9

II Percentage Points of the t Distribution^a

$\nu \backslash \alpha$	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	127.32	318.31	636.62
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	14.089	23.326	31.598
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	7.453	10.213	12.924
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	0.265	0.727	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.019	4.785	5.408
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	0.258	0.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	0.256	0.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.767
24	0.256	0.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	0.256	0.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	0.256	0.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	0.256	0.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	0.256	0.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	0.256	0.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	0.256	0.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	0.255	0.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
60	0.254	0.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
120	0.254	0.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
∞	0.253	0.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

 ν = Degrees of freedom.^aAdapted with permission from *Biometrika Tables for Statisticians*, Vol. 1, 3rd edition, by E. S. Pearson and H. O. Hartley, Cambridge University Press, Cambridge, 1966.

III Percentage Points of the χ^2 Distribution^a

$\nu \backslash \alpha$	0.995	0.990	0.975	0.950	0.500	0.050	0.025	0.010	0.005
1	0.00 +	0.00 +	0.00 +	0.00 +	0.45	3.84	5.02	6.63	7.88
2	0.01	0.02	0.05	0.10	1.39	5.99	7.38	9.21	10.60
3	0.07	0.11	0.22	0.35	2.37	7.81	9.35	11.34	12.84
4	0.21	0.30	0.48	0.71	3.36	9.49	11.14	13.28	14.86
5	0.41	0.55	0.83	1.15	4.35	11.07	12.38	15.09	16.75
6	0.68	0.87	1.24	1.64	5.35	12.59	14.45	16.81	18.55
7	0.99	1.24	1.69	2.17	6.35	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	7.34	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	8.34	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	9.34	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	10.34	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	11.34	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	12.34	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	13.34	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	14.34	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	15.34	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	16.34	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	17.34	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	18.34	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	19.34	31.41	34.17	37.57	40.00
25	10.52	11.52	13.12	14.61	24.34	37.65	40.65	44.31	46.93
30	13.79	14.95	16.79	18.49	29.34	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	39.34	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	49.33	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	59.33	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	69.33	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	79.33	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	89.33	113.14	118.14	124.12	128.30
100	67.33	70.06	74.22	77.93	99.33	124.34	129.56	135.81	140.17

 ν = Degrees of freedom.^aAdapted with permission from *Biometrika Tables for Statisticians*, Vol. 1, 3rd edition by E. S. Pearson and H. O. Hartley, Cambridge University Press, Cambridge, 1966.

$\nu_1 \backslash \nu_2$		Degrees of Freedom for the Numerator (ν_1)																		
		1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
Degrees of Freedom for the Denominator (ν_2)	1	5.83	7.50	8.20	8.58	8.82	8.98	9.10	9.19	9.26	9.32	9.41	9.49	9.58	9.63	9.67	9.71	9.76	9.80	9.85
	2	2.57	3.00	3.15	3.23	3.28	3.31	3.34	3.35	3.37	3.38	3.39	3.41	3.43	3.43	3.44	3.45	3.46	3.47	3.48
	3	2.02	2.28	2.36	2.39	2.41	2.42	2.43	2.44	2.44	2.44	2.45	2.46	2.46	2.46	2.47	2.47	2.47	2.47	2.47
	4	1.81	2.00	2.05	2.06	2.07	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08
	5	1.69	1.85	1.88	1.89	1.89	1.89	1.89	1.89	1.89	1.89	1.89	1.89	1.88	1.88	1.88	1.88	1.87	1.87	1.87
	6	1.62	1.76	1.78	1.79	1.79	1.78	1.78	1.78	1.77	1.77	1.77	1.76	1.76	1.75	1.75	1.75	1.74	1.74	1.74
	7	1.57	1.70	1.72	1.72	1.71	1.71	1.70	1.70	1.70	1.69	1.68	1.68	1.67	1.67	1.66	1.66	1.65	1.65	1.65
	8	1.54	1.66	1.67	1.66	1.66	1.65	1.64	1.64	1.63	1.63	1.62	1.62	1.61	1.60	1.60	1.59	1.59	1.58	1.58
	9	1.51	1.62	1.63	1.63	1.62	1.61	1.60	1.60	1.59	1.59	1.58	1.57	1.56	1.56	1.55	1.54	1.54	1.53	1.53
	10	1.49	1.60	1.60	1.59	1.59	1.58	1.57	1.56	1.56	1.55	1.54	1.53	1.52	1.52	1.51	1.51	1.50	1.49	1.48
	11	1.47	1.58	1.58	1.57	1.56	1.55	1.54	1.53	1.53	1.52	1.51	1.50	1.49	1.49	1.48	1.47	1.47	1.46	1.45
	12	1.46	1.56	1.56	1.55	1.54	1.53	1.52	1.51	1.51	1.50	1.49	1.48	1.47	1.46	1.45	1.45	1.44	1.43	1.42
	13	1.45	1.55	1.55	1.53	1.52	1.51	1.50	1.49	1.49	1.48	1.47	1.46	1.45	1.44	1.43	1.42	1.42	1.41	1.40
	14	1.44	1.53	1.53	1.52	1.51	1.50	1.49	1.48	1.47	1.46	1.45	1.44	1.43	1.42	1.41	1.41	1.40	1.39	1.38
	15	1.43	1.52	1.52	1.51	1.49	1.48	1.47	1.46	1.46	1.45	1.44	1.43	1.41	1.41	1.40	1.39	1.38	1.37	1.36
	16	1.42	1.51	1.51	1.50	1.48	1.47	1.46	1.45	1.44	1.44	1.43	1.41	1.40	1.39	1.38	1.37	1.36	1.35	1.34
	17	1.42	1.51	1.50	1.49	1.47	1.46	1.45	1.44	1.43	1.43	1.41	1.40	1.39	1.38	1.37	1.36	1.35	1.34	1.33
	18	1.41	1.50	1.49	1.48	1.46	1.45	1.44	1.43	1.42	1.42	1.40	1.39	1.38	1.37	1.36	1.35	1.34	1.33	1.32
	19	1.41	1.49	1.49	1.47	1.46	1.44	1.43	1.42	1.41	1.41	1.40	1.38	1.37	1.36	1.35	1.34	1.33	1.32	1.30
	20	1.40	1.49	1.48	1.47	1.45	1.44	1.43	1.42	1.41	1.40	1.39	1.37	1.36	1.35	1.34	1.33	1.32	1.31	1.29
	21	1.40	1.48	1.48	1.46	1.44	1.43	1.42	1.41	1.40	1.39	1.38	1.37	1.35	1.34	1.33	1.32	1.31	1.30	1.28
	22	1.40	1.48	1.47	1.45	1.44	1.42	1.41	1.40	1.39	1.39	1.37	1.36	1.34	1.33	1.32	1.31	1.30	1.29	1.28
	23	1.39	1.47	1.47	1.45	1.43	1.42	1.41	1.40	1.39	1.38	1.37	1.35	1.34	1.33	1.32	1.31	1.30	1.28	1.27
	24	1.39	1.47	1.46	1.44	1.43	1.41	1.40	1.39	1.38	1.38	1.36	1.35	1.33	1.32	1.31	1.30	1.29	1.28	1.26
	25	1.39	1.47	1.46	1.44	1.42	1.41	1.40	1.39	1.38	1.37	1.36	1.34	1.33	1.32	1.31	1.29	1.28	1.27	1.25
	26	1.38	1.46	1.45	1.44	1.42	1.41	1.39	1.38	1.37	1.37	1.35	1.34	1.32	1.31	1.30	1.29	1.28	1.26	1.25
	27	1.38	1.46	1.45	1.43	1.42	1.40	1.39	1.38	1.37	1.36	1.35	1.33	1.32	1.31	1.30	1.28	1.27	1.26	1.24
	28	1.38	1.46	1.45	1.43	1.41	1.40	1.39	1.38	1.37	1.36	1.34	1.33	1.31	1.30	1.29	1.28	1.27	1.25	1.24
	29	1.38	1.45	1.45	1.43	1.41	1.40	1.38	1.37	1.36	1.35	1.34	1.32	1.31	1.30	1.29	1.27	1.26	1.25	1.23
	30	1.38	1.45	1.44	1.42	1.41	1.39	1.38	1.37	1.36	1.35	1.34	1.32	1.30	1.29	1.28	1.27	1.26	1.24	1.23
	40	1.36	1.44	1.42	1.40	1.39	1.37	1.36	1.35	1.34	1.33	1.31	1.30	1.28	1.26	1.25	1.24	1.22	1.21	1.19
	60	1.35	1.42	1.41	1.38	1.37	1.35	1.33	1.32	1.31	1.30	1.29	1.27	1.25	1.24	1.22	1.21	1.19	1.17	1.15
	120	1.34	1.40	1.39	1.37	1.35	1.33	1.31	1.30	1.29	1.28	1.26	1.24	1.22	1.21	1.19	1.18	1.16	1.13	1.10
	∞	1.32	1.39	1.37	1.35	1.33	1.31	1.29	1.28	1.27	1.25	1.24	1.22	1.19	1.18	1.16	1.14	1.12	1.08	1.00

ν = Degrees of freedom.

^aAdapted with permission from *Biometrika Tables for Statisticians*, Vol. 1, 3rd edition by E. S. Pearson and H. O. Hartley, Cambridge University Press, Cambridge, 1966.

IV Percentage Points of the F Distribution (*Continued*)

$$F_{0.10, \nu_1, \nu_2}$$

$\nu_1 \backslash \nu_2$		Degrees of Freedom for the Numerator (ν_1)																		
		1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
Degrees of Freedom for the Denominator (ν_2)	1	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86	60.19	60.71	61.22	61.74	62.00	62.26	62.53	62.79	63.06	63.33
	2	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.48	9.49
	3	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23	5.22	5.20	5.18	5.18	5.17	5.16	5.15	5.14	5.13
	4	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94	3.92	3.90	3.87	3.84	3.83	3.82	3.80	3.79	3.78	3.76
	5	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30	3.27	3.24	3.21	3.19	3.17	3.16	3.14	3.12	3.10
	6	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94	2.90	2.87	2.84	2.82	2.80	2.78	2.76	2.74	2.72
	7	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72	2.70	2.67	2.63	2.59	2.58	2.56	2.54	2.51	2.49	2.47
	8	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.34	2.32	2.29
	9	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42	2.38	2.34	2.30	2.28	2.25	2.23	2.21	2.18	2.16
	10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32	2.28	2.24	2.20	2.18	2.16	2.13	2.11	2.08	2.06
	11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.03	2.00	1.97
	12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19	2.15	2.10	2.06	2.04	2.01	1.99	1.96	1.93	1.90
	13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.90	1.88	1.85
	14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10	2.05	2.01	1.96	1.94	1.91	1.89	1.86	1.83	1.80
	15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06	2.02	1.97	1.92	1.90	1.87	1.85	1.82	1.79	1.76
	16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03	1.99	1.94	1.89	1.87	1.84	1.81	1.78	1.75	1.72
	17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00	1.96	1.91	1.86	1.84	1.81	1.78	1.75	1.72	1.69
	18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98	1.93	1.89	1.84	1.81	1.78	1.75	1.72	1.69	1.66
	19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96	1.91	1.86	1.81	1.79	1.76	1.73	1.70	1.67	1.63
	20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94	1.89	1.84	1.79	1.77	1.74	1.71	1.68	1.64	1.61
	21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92	1.87	1.83	1.78	1.75	1.72	1.69	1.66	1.62	1.59
	22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90	1.86	1.81	1.76	1.73	1.70	1.67	1.64	1.60	1.57
	23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.96	1.92	1.89	1.84	1.80	1.74	1.72	1.69	1.66	1.62	1.59	1.55
	24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88	1.83	1.78	1.73	1.70	1.67	1.64	1.61	1.57	1.53
	25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87	1.82	1.77	1.72	1.69	1.66	1.63	1.59	1.56	1.52
	26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86	1.81	1.76	1.71	1.68	1.65	1.61	1.58	1.54	1.50
	27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85	1.80	1.75	1.70	1.67	1.64	1.60	1.57	1.53	1.49
	28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84	1.79	1.74	1.69	1.66	1.63	1.59	1.56	1.52	1.48
	29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83	1.78	1.73	1.68	1.65	1.62	1.58	1.55	1.51	1.47
	30	2.88	2.49	2.28	2.14	2.03	1.98	1.93	1.88	1.85	1.82	1.77	1.72	1.67	1.64	1.61	1.57	1.54	1.50	1.46
	40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76	1.71	1.66	1.61	1.57	1.54	1.51	1.47	1.42	1.38
	60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71	1.66	1.60	1.54	1.51	1.48	1.44	1.40	1.35	1.29
	120	2.75	2.35	2.13	1.99	1.90	1.82	1.77	1.72	1.68	1.65	1.60	1.55	1.48	1.45	1.41	1.37	1.32	1.26	1.19
	∞	2.71	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60	1.55	1.49	1.42	1.38	1.34	1.30	1.24	1.17	1.00

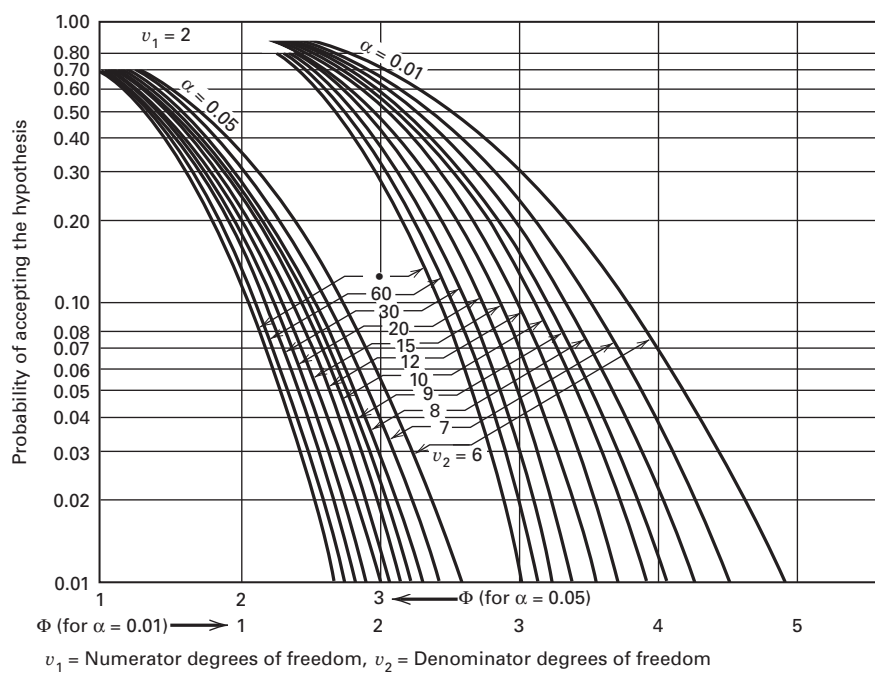
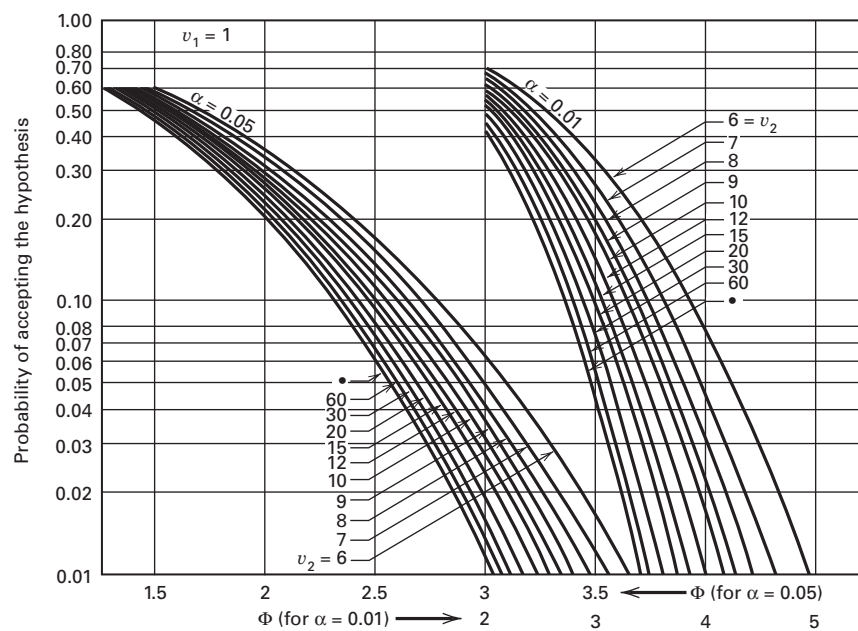
ν_2	ν_1	Degrees of Freedom for the Numerator (ν_1)																		
		1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
Degrees of Freedom for the Denominator (ν_2)	2	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
	3	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
	4	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
	5	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
	6	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
	7	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
	8	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
	9	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
	10	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
	11	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
	12	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
	13	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
	14	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
	15	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
	16	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
	17	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
	18	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
	19	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
	20	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
	21	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
	22	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
	23	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
	24	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
	25	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
	26	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
	27	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
	28	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
	29	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
	30	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
	40	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
	60	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
	120	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
	∞	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.55	1.43	1.35	1.25
		3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

IV Percentage Points of the F Distribution (*Continued*)

$$F_{0.025, \nu_1, \nu_2}$$

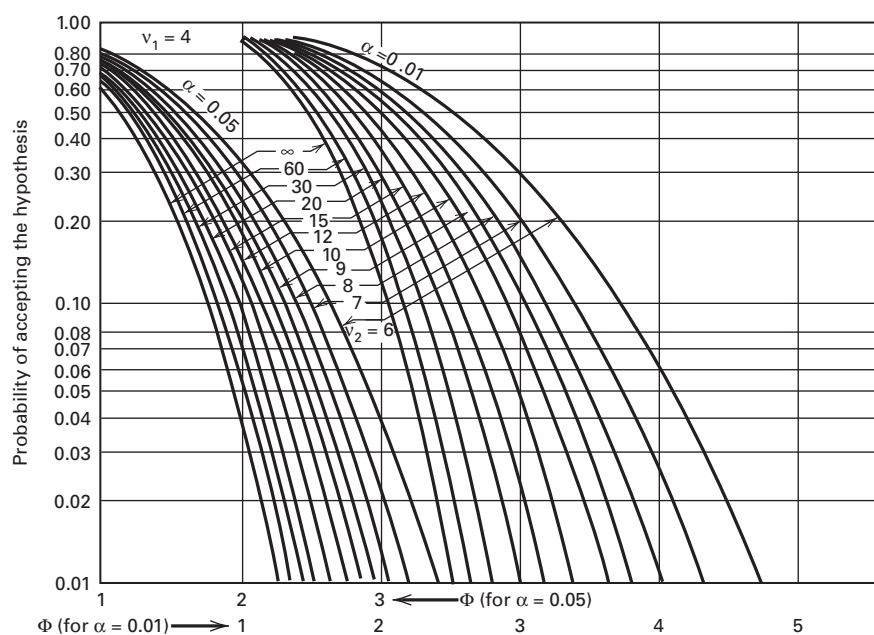
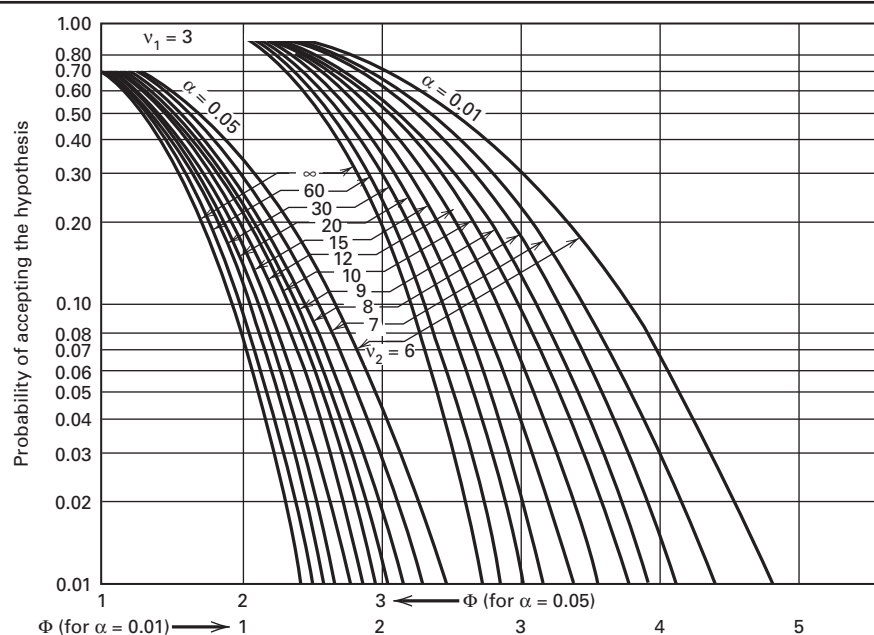
ν_2	ν_1	Degrees of Freedom for the Numerator (ν_1)																		
		1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
Degrees of Freedom for the Denominator (ν_2)	1	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3	968.6	976.7	984.9	993.1	997.2	1001	1006	1010	1014	1018
	2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.49	39.50
	3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.34	14.25	14.17	14.12	14.08	14.04	13.99	13.95	13.90
	4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.75	8.66	8.56	8.51	8.46	8.41	8.36	8.31	8.26
	5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.52	6.43	6.33	6.28	6.23	6.18	6.12	6.07	6.02
	6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.37	5.27	5.17	5.12	5.07	5.01	4.96	4.90	4.85
	7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.67	4.57	4.47	4.42	4.36	4.31	4.25	4.20	4.14
	8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.67
	9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.33
	10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.62	3.52	3.42	3.37	3.31	3.26	3.20	3.14	3.08
	11	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53	3.43	3.33	3.23	3.17	3.12	3.06	3.00	2.94	2.88
	12	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37	3.28	3.18	3.07	3.02	2.96	2.91	2.85	2.79	2.72
	13	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25	3.15	3.05	2.95	2.89	2.84	2.78	2.72	2.66	2.60
	14	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15	3.05	2.95	2.84	2.79	2.73	2.67	2.61	2.55	2.49
	15	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06	2.96	2.86	2.76	2.70	2.64	2.59	2.52	2.46	2.40
	16	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99	2.89	2.79	2.68	2.63	2.57	2.51	2.45	2.38	2.32
	17	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92	2.82	2.72	2.62	2.56	2.50	2.44	2.38	2.32	2.25
	18	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87	2.77	2.67	2.56	2.50	2.44	2.38	2.32	2.26	2.19
	19	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82	2.72	2.62	2.51	2.45	2.39	2.33	2.27	2.20	2.13
	20	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77	2.68	2.57	2.46	2.41	2.35	2.29	2.22	2.16	2.09
	21	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80	2.73	2.64	2.53	2.42	2.37	2.31	2.25	2.18	2.11	2.04
	22	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76	2.70	2.60	2.50	2.39	2.33	2.27	2.21	2.14	2.08	2.00
	23	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73	2.67	2.57	2.47	2.36	2.30	2.24	2.18	2.11	2.04	1.97
	24	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64	2.54	2.44	2.33	2.27	2.21	2.15	2.08	2.01	1.94
	25	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61	2.51	2.41	2.30	2.24	2.18	2.12	2.05	1.98	1.91
	26	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.73	2.65	2.59	2.49	2.39	2.28	2.22	2.16	2.09	2.03	1.95	1.88
	27	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.71	2.63	2.57	2.47	2.36	2.25	2.19	2.13	2.07	2.00	1.93	1.85
	28	5.61	4.22	3.63	3.29	3.06	2.90	2.78	2.69	2.61	2.55	2.45	2.34	2.23	2.17	2.11	2.05	1.98	1.91	1.83
	29	5.59	4.20	1.61	3.27	3.04	2.88	2.76	2.67	2.59	2.53	2.43	2.32	2.21	2.15	2.09	2.03	1.96	1.89	1.81
	30	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51	2.41	2.31	2.20	2.14	2.07	2.01	1.94	1.87	1.79
	40	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39	2.29	2.18	2.07	2.01	1.94	1.88	1.80	1.72	1.64
	60	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27	2.17	2.06	1.94	1.88	1.82	1.74	1.67	1.58	1.48
	120	5.15	3.80	3.23	2.89	2.67	2.52	2.39	2.30	2.22	2.16	2.05	1.94	1.82	1.76	1.69	1.61	1.53	1.43	1.31
	∞	5.02	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05	1.94	1.83	1.71	1.64	1.57	1.48	1.39	1.27	1.00

ν_2	ν_1	Degrees of Freedom for the Numerator (ν_1)																		
		1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
Degrees of Freedom for the Denominator (ν_2)	1	4052	4999.5	5403	5625	5764	5859	5928	5982	6022	6056	6106	6157	6209	6235	6261	6287	6313	6339	6366
	2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50
	3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.87	26.69	26.00	26.50	26.41	26.32	26.22	26.13
	4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46
	5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
	6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
	7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
	8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86
	9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31
	10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91
	11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60
	12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
	13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17
	14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00
	15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
	16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75
	17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65
	18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57
	19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49
	20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42
	21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.36
	22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
	23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
	24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
	25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
	26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	2.96	2.81	2.66	2.58	2.50	2.42	2.33	2.23	2.13
	27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
	28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.90	2.75	2.60	2.52	2.44	2.34	2.26	2.17	2.06
	29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
	30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
	40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
	60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
	120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53	1.38
	∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

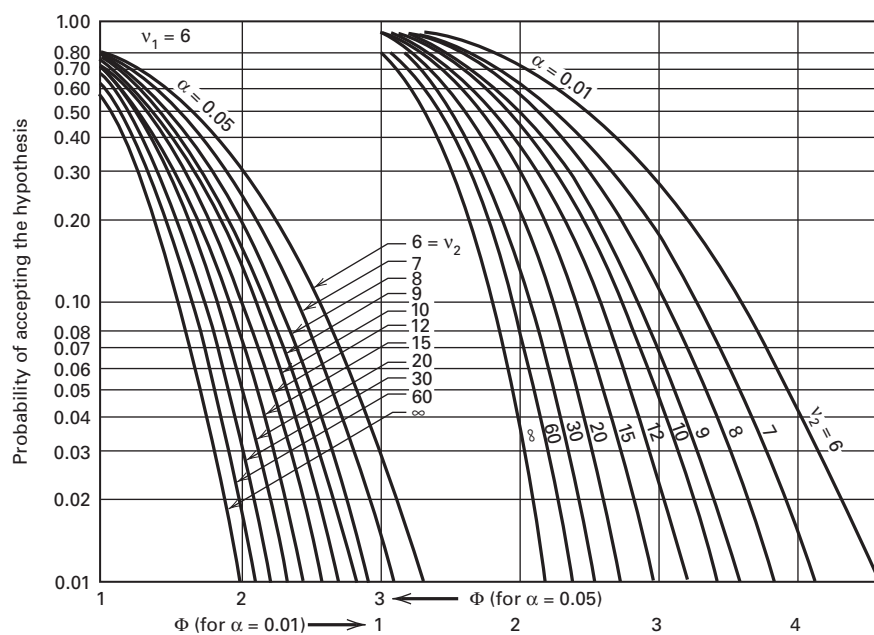
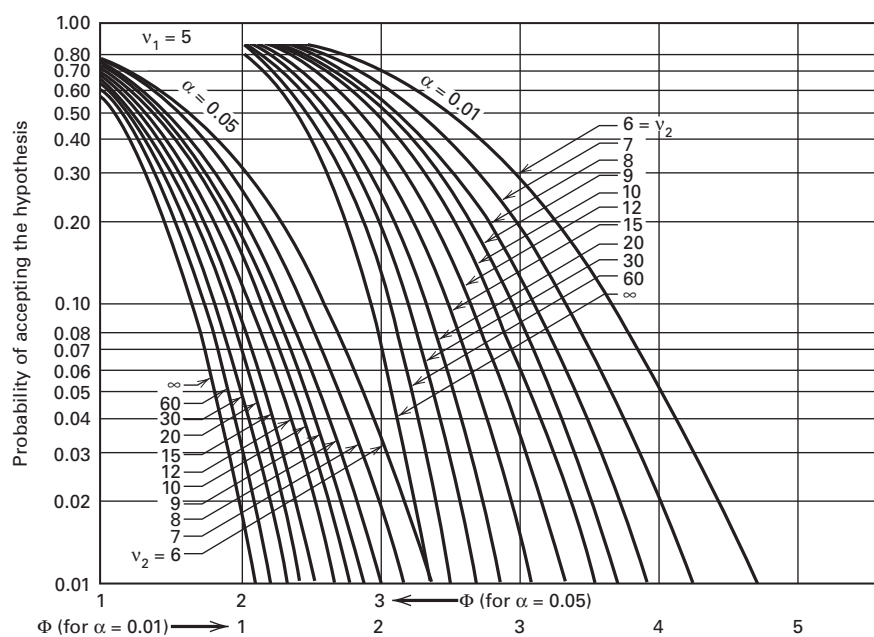
V Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance^a

^aAdapted with permission from *Biometrika Tables for Statisticians*, Vol. 2, by E. S. Pearson and H. O. Hartley, Cambridge University Press, Cambridge, 1972.

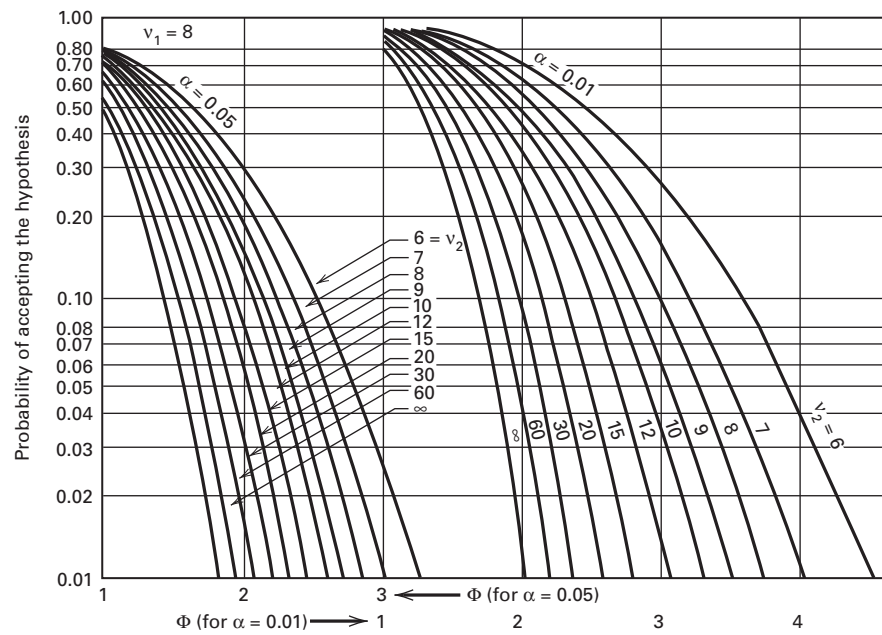
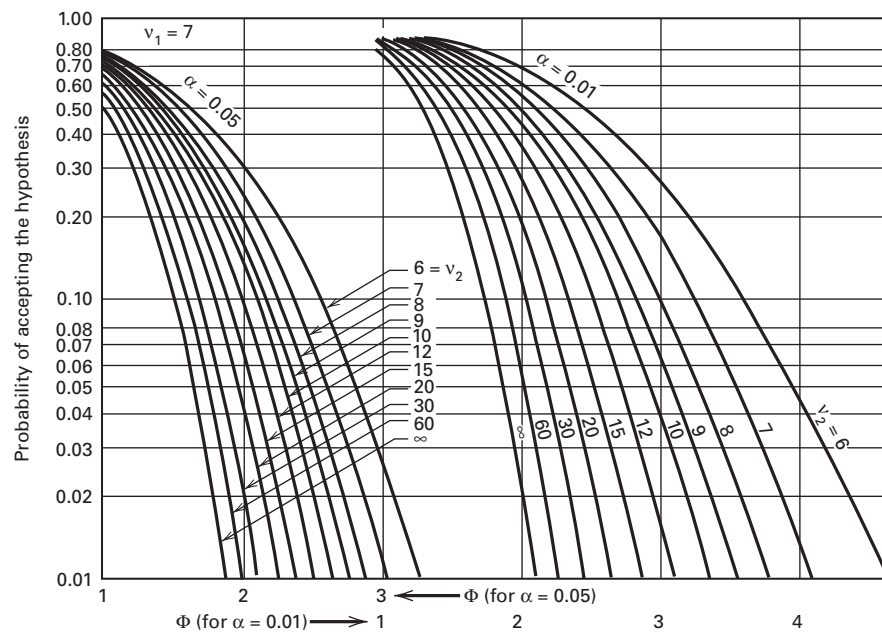
V Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance (Continued)



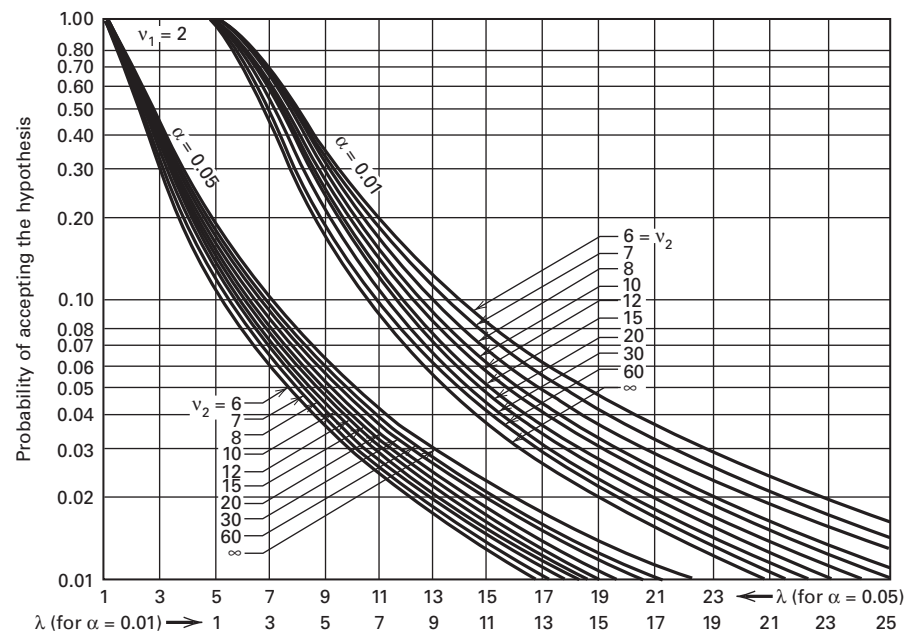
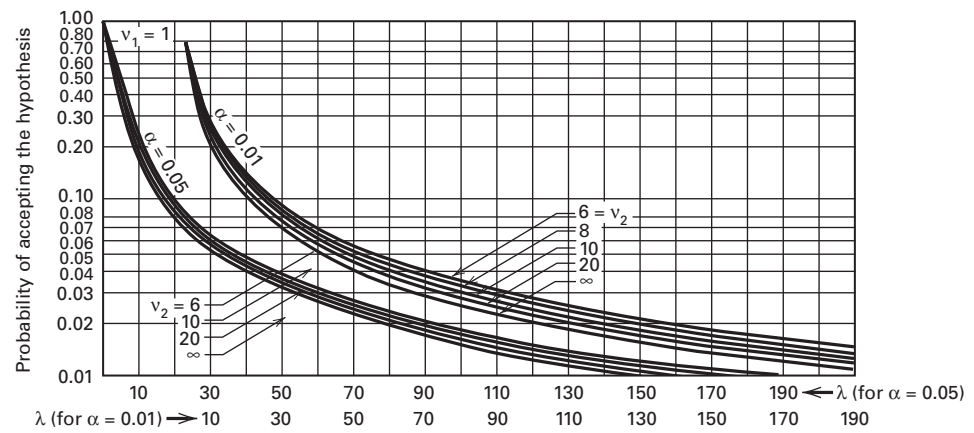
V Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance (Continued)



V Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance (Continued)

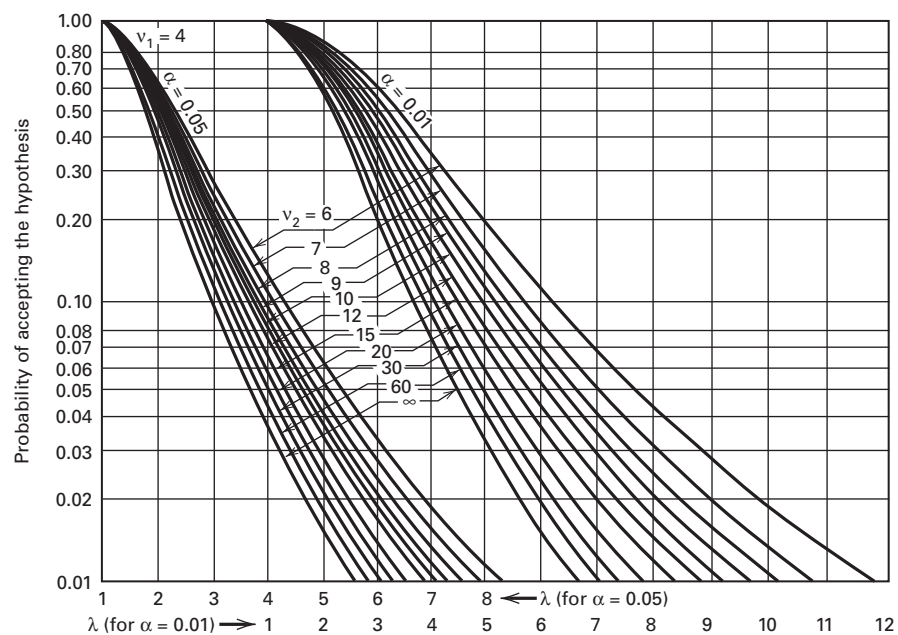
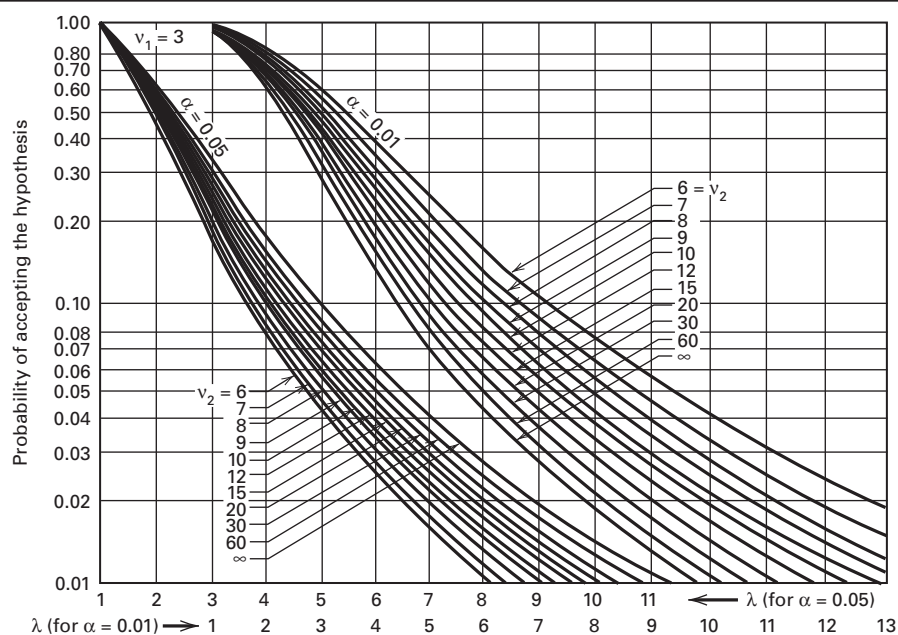


VI Operating Characteristic Curves for the Random Effects Model Analysis of Variance^a

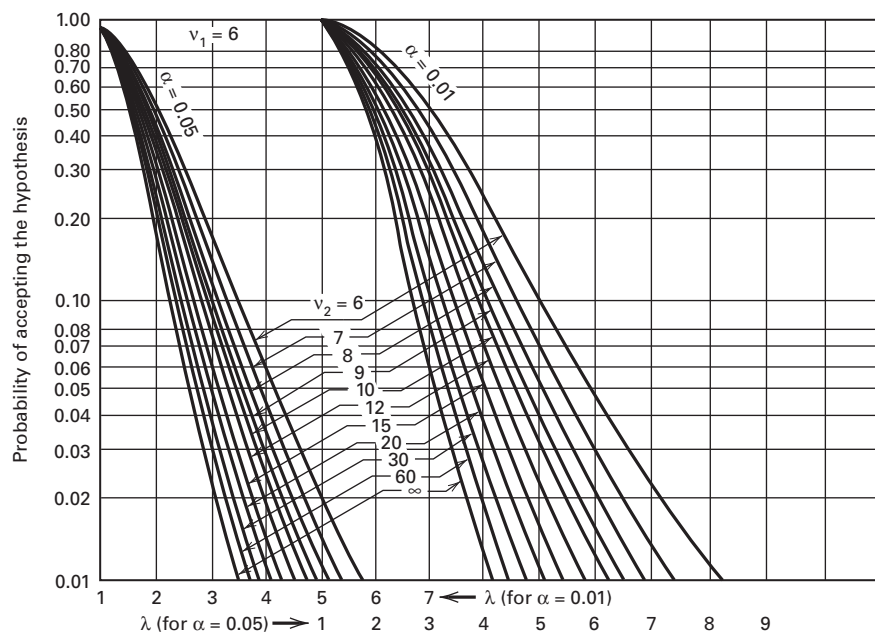
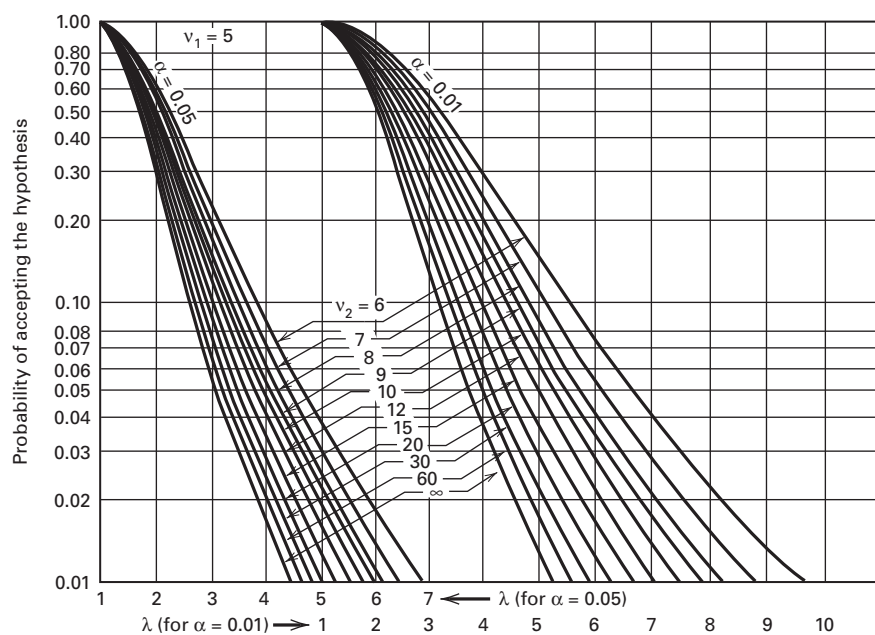


^aReproduced with permission from *Engineering Statistics*, 2nd edition, by A. H. Bowker and G. J. Lieberman, Prentice Hall, Inc., Englewood Cliffs, N.J., 1972.

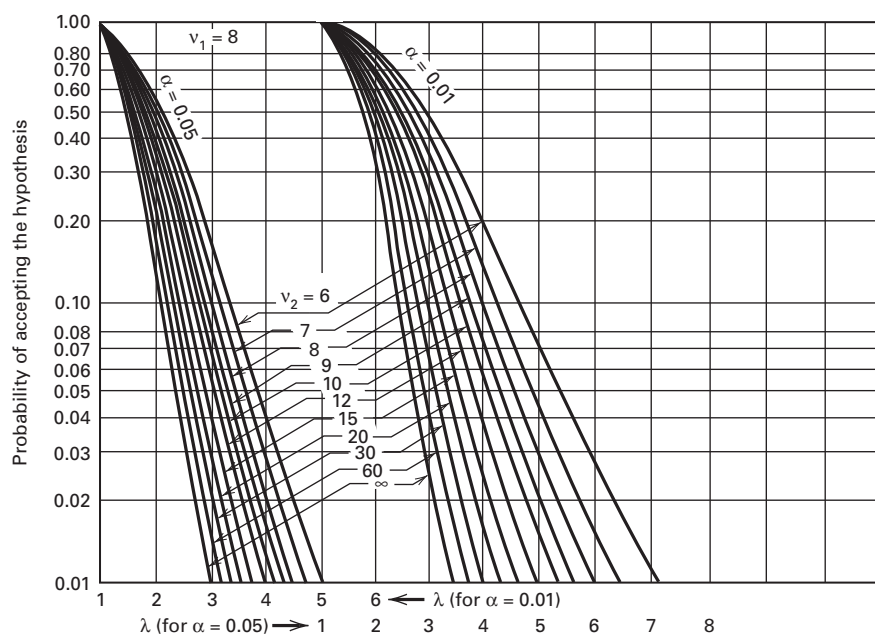
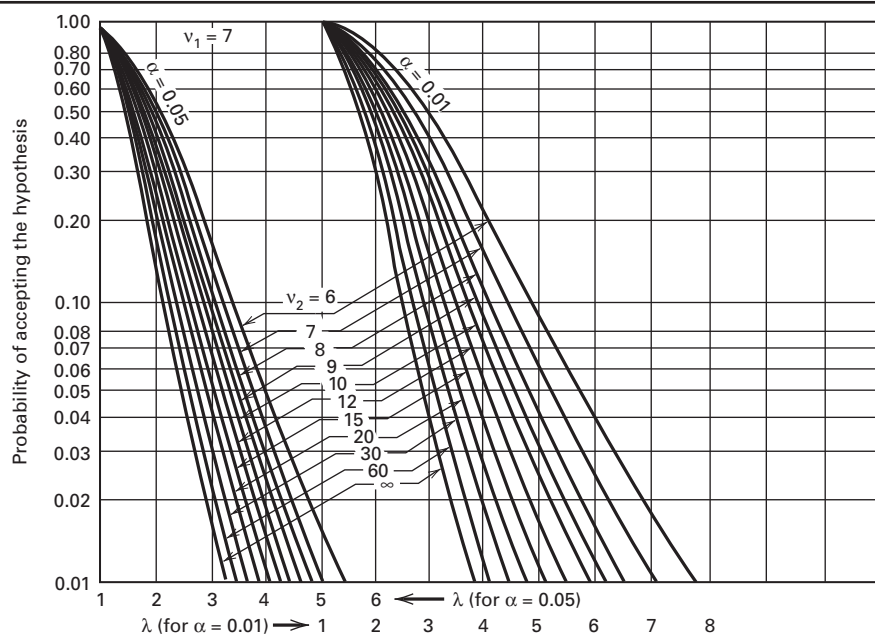
VI Operating Characteristic Curves for the Random Effects Model Analysis of Variance (Continued)



VI Operating Characteristic Curves for the Random Effects Model Analysis of Variance (Continued)



VI Operating Characteristic Curves for the Random Effects Model Analysis of Variance (*Continued*)



VII Percentage Points of the Studentized Range Statistic^a

$$q_{0.01}(p, f)$$

<i>f</i>	<i>p</i>																		
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	90	135	164	186	202	216	227	237	246	253	260	266	272	272	282	286	290	294	298
2	14.0	19.0	22.3	24.7	26.6	28.2	29.5	30.7	31.7	32.6	33.4	31.4	34.8	35.4	36.0	36.5	37.0	37.5	37.9
3	8.26	10.6	12.2	13.3	14.2	15.0	15.6	16.2	16.7	17.1	17.5	17.9	18.2	18.5	18.8	19.1	19.3	19.5	19.8
4	6.51	8.12	9.17	9.96	10.6	11.1	11.5	11.9	12.3	12.6	12.8	13.1	13.3	13.5	13.7	13.9	14.1	14.2	14.4
5	5.70	6.97	7.80	8.42	8.91	9.32	9.67	9.97	10.24	10.48	10.70	10.89	11.08	11.24	11.40	11.55	11.68	11.81	11.93
6	5.24	6.33	7.03	7.56	7.97	8.32	8.61	8.87	9.10	9.30	9.49	9.65	9.81	9.95	10.08	10.21	10.32	10.43	10.54
7	4.95	5.92	6.54	7.01	7.37	7.68	7.94	8.17	8.37	8.55	8.71	8.86	9.00	9.12	9.24	9.35	9.46	9.55	9.65
8	4.74	5.63	6.20	6.63	6.96	7.24	7.47	7.68	7.87	8.03	8.18	8.31	8.44	8.55	8.66	8.76	8.85	8.94	9.03
9	4.60	5.43	5.96	6.35	6.66	6.91	7.13	7.32	7.49	7.65	7.78	7.91	8.03	8.13	8.23	8.32	8.41	8.49	8.57
10	4.48	5.27	5.77	6.14	6.43	6.67	6.87	7.05	7.21	7.36	7.48	7.60	7.71	7.81	7.91	7.99	8.07	8.15	8.22
11	4.39	5.14	5.62	5.97	6.25	6.48	6.67	6.84	6.99	7.13	7.25	7.36	7.46	7.56	7.65	7.73	7.81	7.88	7.95
12	4.32	5.04	5.50	5.84	6.10	6.32	6.51	6.67	6.81	6.94	7.06	7.17	7.26	7.36	7.44	7.52	7.59	7.66	7.73
13	4.26	4.96	5.40	5.73	5.98	6.19	6.37	6.53	6.67	6.79	6.90	7.01	7.10	7.19	7.27	7.34	7.42	7.48	7.55
14	4.21	4.89	5.32	5.63	5.88	6.08	6.26	6.41	6.54	6.66	6.77	6.87	6.96	7.05	7.12	7.20	7.27	7.33	7.39
15	4.17	4.83	5.25	5.56	5.80	5.99	6.16	6.31	6.44	6.55	6.66	6.76	6.84	6.93	7.00	7.07	7.14	7.20	7.26
16	4.13	4.78	5.19	5.49	5.72	5.92	6.08	6.22	6.35	6.46	6.56	6.66	6.74	6.82	6.90	6.97	7.03	7.09	7.15
17	4.10	4.74	5.14	5.43	5.66	5.85	6.01	6.15	6.27	6.38	6.48	6.57	6.66	6.73	6.80	6.87	6.94	7.00	7.05
18	4.07	4.70	5.09	5.38	5.60	5.79	5.94	6.08	6.20	6.31	6.41	6.50	6.58	6.65	6.72	6.79	6.85	6.91	6.96
19	4.05	4.67	5.05	5.33	5.55	5.73	5.89	6.02	6.14	6.25	6.34	6.43	6.51	6.58	6.65	6.72	6.78	6.84	6.89
20	4.02	4.64	5.02	5.29	5.51	5.69	5.84	5.97	6.09	6.19	6.29	6.37	6.45	6.52	6.59	6.65	6.71	6.76	6.82
24	3.96	4.54	4.91	5.17	5.37	5.54	5.69	5.81	5.92	6.02	6.11	6.19	6.26	6.33	6.39	6.45	6.51	6.56	6.61
30	3.89	4.45	4.80	5.05	5.24	5.40	5.54	5.65	5.76	5.85	5.93	6.01	6.08	6.14	6.20	6.26	6.31	6.36	6.41
40	3.82	4.37	4.70	4.93	5.11	5.27	5.39	5.50	5.60	5.69	5.77	5.84	5.90	5.96	6.02	6.07	6.12	6.17	6.21
60	3.76	4.28	4.60	4.82	4.99	5.13	5.25	5.36	5.45	5.53	5.60	5.67	5.73	5.79	5.84	5.89	5.93	5.98	6.02
120	3.70	4.20	4.50	4.71	4.87	5.01	5.12	5.21	5.30	5.38	5.44	5.51	5.56	5.61	5.66	5.71	5.75	5.79	5.83
∞	3.64	4.12	4.40	4.60	4.76	4.88	4.99	5.08	5.16	5.23	5.29	5.35	5.40	5.45	5.49	5.54	5.57	5.61	5.65

f = Degrees of freedom.

^aFrom J. M. May, "Extended and Corrected Tables of the Upper Percentage Points of the Studentized Range," *Biometrika*, Vol. 39, pp. 192–193, 1952. Reproduced by permission of the trustees of *Biometrika*.

p

f	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	18.1	26.7	32.8	37.2	40.5	43.1	45.4	47.3	49.1	50.6	51.9	53.2	54.3	55.4	56.3	57.2	58.0	58.8	59.6
2	6.09	8.28	9.80	10.89	11.73	12.43	13.03	13.54	13.99	14.39	14.75	15.08	15.38	15.65	15.91	16.14	16.36	16.57	16.77
3	4.50	5.88	6.83	7.51	8.04	8.47	8.85	9.18	9.46	9.72	9.95	10.16	10.35	10.52	10.69	10.84	10.98	11.12	11.24
4	3.93	5.00	5.76	6.31	6.73	7.06	7.35	7.60	7.83	8.03	8.21	8.37	8.52	8.67	8.80	8.92	9.03	9.14	9.24
5	3.64	4.60	5.22	5.67	6.03	6.33	6.58	6.80	6.99	7.17	7.32	7.47	7.60	7.72	7.83	7.93	8.03	8.12	8.21
6	3.46	4.34	4.90	5.31	5.63	5.89	6.12	6.32	6.49	6.65	6.79	6.92	7.04	7.14	7.24	7.34	7.43	7.51	7.59
7	3.34	4.16	4.68	5.06	5.35	5.59	5.80	5.99	6.15	6.29	6.42	6.54	6.65	6.75	6.84	6.93	7.01	7.08	7.16
8	3.26	4.04	4.53	4.89	5.17	5.40	5.60	5.77	5.92	6.05	6.18	6.29	6.39	6.48	6.57	6.65	6.73	6.80	6.87
9	3.20	3.95	4.42	4.76	5.02	5.24	5.43	5.60	5.74	5.87	5.98	6.09	6.19	6.28	6.36	6.44	6.51	6.58	6.65
10	3.15	3.88	4.33	4.66	4.91	5.12	5.30	5.46	5.60	5.72	5.83	5.93	6.03	6.12	6.20	6.27	6.34	6.41	6.47
11	3.11	3.82	4.26	4.58	4.82	5.03	5.20	5.35	5.49	5.61	5.71	5.81	5.90	5.98	6.06	6.14	6.20	6.27	6.33
12	3.08	3.77	4.20	4.51	4.75	4.95	5.12	5.27	5.40	5.51	5.61	5.71	5.80	5.88	5.95	6.02	6.09	6.15	6.21
13	3.06	3.73	4.15	4.46	4.69	4.88	5.05	5.19	5.32	5.43	5.53	5.63	5.71	5.79	5.86	5.93	6.00	6.06	6.11
14	3.03	3.70	4.11	4.41	4.64	4.83	4.99	5.13	5.25	5.36	5.46	5.56	5.64	5.72	5.79	5.86	5.92	5.98	6.03
15	3.01	3.67	4.08	4.37	4.59	4.78	4.94	5.08	5.20	5.31	5.40	5.49	5.57	5.65	5.72	5.79	5.85	5.91	5.96
16	3.00	3.65	4.05	4.34	4.56	4.74	4.90	5.03	5.15	5.26	5.35	5.44	5.52	5.59	5.66	5.73	5.79	5.84	5.90
17	2.98	3.62	4.02	4.31	4.52	4.70	4.86	4.99	5.11	5.21	5.31	5.39	5.47	5.55	5.61	5.68	5.74	5.79	5.84
18	2.97	3.61	4.00	4.28	4.49	4.67	4.83	4.96	5.07	5.17	5.27	5.35	5.43	5.50	5.57	5.63	5.69	5.74	5.79
19	2.96	3.59	3.98	4.26	4.47	4.64	4.79	4.92	5.04	5.14	5.23	5.32	5.39	5.46	5.53	5.59	5.65	5.70	5.75
20	2.95	3.58	3.96	4.24	4.45	4.62	4.77	4.90	5.01	5.11	5.20	5.28	5.36	5.43	5.50	5.56	5.61	5.66	5.71
24	2.92	3.53	3.90	4.17	4.37	4.54	4.68	4.81	4.92	5.01	5.10	5.18	5.25	5.32	5.38	5.44	5.50	5.55	5.59
30	2.89	3.48	3.84	4.11	4.30	4.46	4.60	4.72	4.83	4.92	5.00	5.08	5.15	5.21	5.27	5.33	5.38	5.43	5.48
40	2.86	3.44	3.79	4.04	4.23	4.39	4.52	4.63	4.74	4.82	4.90	4.98	5.05	5.11	5.17	5.22	5.27	5.32	5.36
60	2.83	3.40	3.74	3.98	4.16	4.31	4.44	4.55	4.65	4.73	4.81	4.88	4.94	5.00	5.06	5.11	5.15	5.20	5.24
120	2.80	3.36	3.69	3.92	4.10	4.24	4.36	4.47	4.56	4.64	4.71	4.78	4.84	4.90	4.95	5.00	5.04	5.09	5.13
∞	2.77	3.32	3.63	3.86	4.03	4.17	4.29	4.39	4.47	4.55	4.62	4.68	4.74	4.80	4.84	4.88	4.93	4.97	5.01

VIII Critical Values for Dunnett's Test for Comparing Treatments with a Control^a

$$d_{0.05}(a - 1, f)$$

Two-Sided Comparisons

f	$a - 1 = \text{Number of Treatment Means (Excluding Control)}$								
	1	2	3	4	5	6	7	8	9
5	2.57	3.03	3.29	3.48	3.62	3.73	3.82	3.90	3.97
6	2.45	2.86	3.10	3.26	3.39	3.49	3.57	3.64	3.71
7	2.36	2.75	2.97	3.12	3.24	3.33	3.41	3.47	3.53
8	2.31	2.67	2.88	3.02	3.13	3.22	3.29	3.35	3.41
9	2.26	2.61	2.81	2.95	3.05	3.14	3.20	3.26	3.32
10	2.23	2.57	2.76	2.89	2.99	3.07	3.14	3.19	3.24
11	2.20	2.53	2.72	2.84	2.94	3.02	3.08	3.14	3.19
12	2.18	2.50	2.68	2.81	2.90	2.98	3.04	3.09	3.14
13	2.16	2.48	2.65	2.78	2.87	2.94	3.00	3.06	3.10
14	2.14	2.46	2.63	2.75	2.84	2.91	2.97	3.02	3.07
15	2.13	2.44	2.61	2.73	2.82	2.89	2.95	3.00	3.04
16	2.12	2.42	2.59	2.71	2.80	2.87	2.92	2.97	3.02
17	2.11	2.41	2.58	2.69	2.78	2.85	2.90	2.95	3.00
18	2.10	2.40	2.56	2.68	2.76	2.83	2.89	2.94	2.98
19	2.09	2.39	2.55	2.66	2.75	2.81	2.87	2.92	2.96
20	2.09	2.38	2.54	2.65	2.73	2.80	2.86	2.90	2.95
24	2.06	2.35	2.51	2.61	2.70	2.76	2.81	2.86	2.90
30	2.04	2.32	2.47	2.58	2.66	2.72	2.77	2.82	2.86
40	2.02	2.29	2.44	2.54	2.62	2.68	2.73	2.77	2.81
60	2.00	2.27	2.41	2.51	2.58	2.64	2.69	2.73	2.77
120	1.98	2.24	2.38	2.47	2.55	2.60	2.65	2.69	2.73
∞	1.96	2.21	2.35	2.44	2.51	2.57	2.61	2.65	2.69

$$d_{0.01}(a - 1, f)$$

Two-Sided Comparisons

f	$a - 1 = \text{Number of Treatment Means (Excluding Control)}$								
	1	2	3	4	5	6	7	8	9
5	4.03	4.63	4.98	5.22	5.41	5.56	5.69	5.80	5.89
6	3.71	4.21	4.51	4.71	4.87	5.00	5.10	5.20	5.28
7	3.50	3.95	4.21	4.39	4.53	4.64	4.74	4.82	4.89
8	3.36	3.77	4.00	4.17	4.29	4.40	4.48	4.56	4.62
9	3.25	3.63	3.85	4.01	4.12	4.22	4.30	4.37	4.43
10	3.17	3.53	3.74	3.88	3.99	4.08	4.16	4.22	4.28
11	3.11	3.45	3.65	3.79	3.89	3.98	4.05	4.11	4.16
12	3.05	3.39	3.58	3.71	3.81	3.89	3.96	4.02	4.07
13	3.01	3.33	3.52	3.65	3.74	3.82	3.89	3.94	3.99
14	2.98	3.29	3.47	3.59	3.69	3.76	3.83	3.88	3.93
15	2.95	3.25	3.43	3.55	3.64	3.71	3.78	3.83	3.88
16	2.92	3.22	3.39	3.51	3.60	3.67	3.73	3.78	3.83
17	2.90	3.19	3.36	3.47	3.56	3.63	3.69	3.73	3.79
18	2.88	3.17	3.33	3.44	3.53	3.60	3.66	3.71	3.75
19	2.86	3.15	3.31	3.42	3.50	3.57	3.63	3.68	3.72
20	2.85	3.13	3.29	3.40	3.48	3.55	3.60	3.65	3.69
24	2.80	3.07	3.22	3.32	3.40	3.47	3.52	3.57	3.61
30	2.75	3.01	3.15	3.25	3.33	3.39	3.44	3.49	3.52
40	2.70	2.95	3.09	3.19	3.26	3.32	3.37	3.41	3.44
60	2.66	2.90	3.03	3.12	3.19	3.25	3.29	3.33	3.37
120	2.62	2.85	2.97	3.06	3.12	3.18	3.22	3.26	3.29
∞	2.58	2.79	2.92	3.00	3.06	3.11	3.15	3.19	3.22

 f = Degrees of freedom.

^aReproduced with permission from C. W. Dunnett, "New Tables for Multiple Comparison with a Control," *Biometrics*, Vol. 20, No. 3, 1964, and from C. W. Dunnett, "A Multiple Comparison Procedure for Comparing Several Treatments with a Control," *Journal of the American Statistical Association*, Vol. 50, 1955.

VIII Critical Values for Dunnett's Test for Comparing Treatments with a Control

(Continued)

$$d_{0.05}(a-1, f)$$

One-Sided Comparisons

f	$a-1 = \text{Number of Treatment Means (Excluding Control)}$								
	1	2	3	4	5	6	7	8	9
5	2.02	2.44	2.68	2.85	2.98	3.08	3.16	3.24	3.30
6	1.94	2.34	2.56	2.71	2.83	2.92	3.00	3.07	3.12
7	1.89	2.27	2.48	2.62	2.73	2.82	2.89	2.95	3.01
8	1.86	2.22	2.42	2.55	2.66	2.74	2.81	2.87	2.92
9	1.83	2.18	2.37	2.50	2.60	2.68	2.75	2.81	2.86
10	1.81	2.15	2.34	2.47	2.56	2.64	2.70	2.76	2.81
11	1.80	2.13	2.31	2.44	2.53	2.60	2.67	2.72	2.77
12	1.78	2.11	2.29	2.41	2.50	2.58	2.64	2.69	2.74
13	1.77	2.09	2.27	2.39	2.48	2.55	2.61	2.66	2.71
14	1.76	2.08	2.25	2.37	2.46	2.53	2.59	2.64	2.69
15	1.75	2.07	2.24	2.36	2.44	2.51	2.57	2.62	2.67
16	1.75	2.06	2.23	2.34	2.43	2.50	2.56	2.61	2.65
17	1.74	2.05	2.22	2.33	2.42	2.49	2.54	2.59	2.64
18	1.73	2.04	2.21	2.32	2.41	2.48	2.53	2.58	2.62
19	1.73	2.03	2.20	2.31	2.40	2.47	2.52	2.57	2.61
20	1.72	2.03	2.19	2.30	2.39	2.46	2.51	2.56	2.60
24	1.71	2.01	2.17	2.28	2.36	2.43	2.48	2.53	2.57
30	1.70	1.99	2.15	2.25	2.33	2.40	2.45	2.50	2.54
40	1.68	1.97	2.13	2.23	2.31	2.37	2.42	2.47	2.51
60	1.67	1.95	2.10	2.21	2.28	2.35	2.39	2.44	2.48
120	1.66	1.93	2.08	2.18	2.26	2.32	2.37	2.41	2.45
∞	1.64	1.92	2.06	2.16	2.23	2.29	2.34	2.38	2.42

$$d_{0.01}(a-1, f)$$

One-Sided Comparisons

f	$a-1 = \text{Number of Treatment Means (Excluding Control)}$								
	1	2	3	4	5	6	7	8	9
5	3.37	3.90	4.21	4.43	4.60	4.73	4.85	4.94	5.03
6	3.14	3.61	3.88	4.07	4.21	4.33	4.43	4.51	4.59
7	3.00	3.42	3.66	3.83	3.96	4.07	4.15	4.23	4.30
8	2.90	3.29	3.51	3.67	3.79	3.88	3.96	4.03	4.09
9	2.82	3.19	3.40	3.55	3.66	3.75	3.82	3.89	3.94
10	2.76	3.11	3.31	3.45	3.56	3.64	3.71	3.78	3.83
11	2.72	3.06	3.25	3.38	3.48	3.56	3.63	3.69	3.74
12	2.68	3.01	3.19	3.32	3.42	3.50	3.56	3.62	3.67
13	2.65	2.97	3.15	3.27	3.37	3.44	3.51	3.56	3.61
14	2.62	2.94	3.11	3.23	3.32	3.40	3.46	3.51	3.56
15	2.60	2.91	3.08	3.20	3.29	3.36	3.42	3.47	3.52
16	2.58	2.88	3.05	3.17	3.26	3.33	3.39	3.44	3.48
17	2.57	2.86	3.03	3.14	3.23	3.30	3.36	3.41	3.45
18	2.55	2.84	3.01	3.12	3.21	3.27	3.33	3.38	3.42
19	2.54	2.83	2.99	3.10	3.18	3.25	3.31	3.36	3.40
20	2.53	2.81	2.97	3.08	3.17	3.23	3.29	3.34	3.38
24	2.49	2.77	2.92	3.03	3.11	3.17	3.22	3.27	3.31
30	2.46	2.72	2.87	2.97	3.05	3.11	3.16	3.21	3.24
40	2.42	2.68	2.82	2.92	2.99	3.05	3.10	3.14	3.18
60	2.39	2.64	2.78	2.87	2.94	3.00	3.04	3.08	3.12
120	2.36	2.60	2.73	2.82	2.89	2.94	2.99	3.03	3.06
∞	2.33	2.56	2.68	2.77	2.84	2.89	2.93	2.97	3.00

IX Coefficients of Orthogonal Polynomials^a

n = 3			n = 4			n = 5			n = 6			n = 7			
X_j	P_1	P_2	P_1	P_2	P_3	P_1	P_2	P_3	P_4	P_1	P_2	P_3	P_4	P_5	P_6
1	-1	1	-3	1	-1	-2	2	-1	1	-5	5	-5	1	-1	3
2	0	-2	-1	-1	3	-1	-1	2	-4	-3	-1	7	-3	5	-6
3	1	1	1	-1	-3	0	-2	0	6	-1	-4	4	2	-10	15
4			3	1	1	1	-1	-2	-4	1	-4	-4	2	10	0
5						2	2	1	1	3	-1	-7	-3	-5	-20
6										5	5				15
7															-6
$\sum_{j=1}^n [P(X_j)]^2$	2	6	20	4	20	10	14	10	70	70	84	180	28	252	924
	λ	1	3	2	$\frac{10}{3}$	1	1	$\frac{5}{6}$	$\frac{35}{12}$	2	$\frac{3}{2}$	$\frac{5}{3}$	$\frac{7}{12}$	$\frac{21}{10}$	$\frac{77}{60}$

n = 8			n = 9			n = 10												
X_j	P_1	P_2	P_3	P_4	P_5	P_6	P_1	P_2	P_3	P_4	P_5	P_6						
1	-7	7	-7	7	-7	1	-4	28	-14	14	-4	4						
2	-5	1	5	-13	23	-5	-3	7	7	-21	11	-17						
3	-3	-3	7	-3	-17	9	-2	-8	13	-11	-4	22						
4	-1	-5	3	9	-15	-5	-1	-17	9	9	-9	1						
5	1	-5	-3	9	15	-5	0	-20	0	18	0	-20						
6	3	-3	-7	-3	17	9	1	-17	-9	9	9	1						
7	5	1	-5	-13	-23	-5	2	-8	-13	-11	4	22						
8	7	7	7	7	7	1	3	7	-7	-21	-11	-17						
9							4	28	14	14	4	4						
10																		
$\sum_{j=1}^n [P(X_j)]^2$	168	168	264	616	2184	264	60	2772	990	2002	468	1980	330	132	8580	2860	780	660
	λ	2	1	$\frac{2}{3}$	$\frac{7}{12}$	$\frac{7}{10}$	1	3	$\frac{5}{6}$	$\frac{7}{12}$	$\frac{3}{20}$	$\frac{11}{60}$	2	$\frac{1}{2}$	$\frac{5}{3}$	$\frac{5}{12}$	$\frac{1}{10}$	$\frac{11}{240}$

^aAdapted with permission from *Biometrika Tables for Statisticians*, Vol. 1, 3rd edition by E. S. Pearson and H. O. Hartley, Cambridge University Press, Cambridge, 1966.

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$

Designs with 3 Factors		
(a) 2^{3-1} ; 1/2 fraction of 3 factors in 4 runs	<u>Design Generators</u> $C = AB$ Defining relation: $I = ABC$ <u>Aliases</u> $A = BC$ $B = AC$ $C = AB$	Resolution III
Designs with 4 Factors		
(b) 2^{4-1} ; 1/2 fraction of 4 factors in 8 runs	<u>Design Generators</u> $D = ABC$ Defining relation: $I = ABCD$ <u>Aliases</u> $A = BCD$ $B = ACD$ $C = ABD$ $D = ABC$ $AB = CD$ $AC = BD$ $AD = BC$	Resolution IV
Designs with 5 Factors		
(c) 2^{5-2} ; 1/4 fraction of 5 factors in 8 runs	<u>Design Generators</u> $D = AB \quad E = AC$ Defining relation: $I = ABD = ACE = BCDE$ <u>Aliases</u> $A = BD = CE$ $B = AD = CDE$ $C = AE = BDE$ $D = AB = BCE$ $E = AC = BCD$ $BC = DE = ACD = ABE$ $CD = BE = ABC = ADE$	Resolution III
(d) 2^{5-1} ; 1/2 fraction of 5 factors in 16 runs	<u>Design Generators</u> $E = ABCD$ Defining relation: $I = ABCDE$ <u>Aliases</u> Each main effect is aliased with a single 4-factor interaction. $AB = CDE \quad BD = ACE$ $AC = BDE \quad BE = ACD$ $AD = BCE \quad CD = ABE$ $AE = BCD \quad CE = ABD$ $BC = ADE \quad DE = ABC$	Resolution V

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 6 Factors

- (e) 2^{6-3} ; 1/8 fraction of
6 factors in 8 runs

Resolution III

Design Generators

$$D = AB \quad E = AC \quad F = BC$$

$$\text{Defining relation: } I = ABD = ACE = BCDE = BCF = ACDF = ABEF = DEF$$

Aliases

$$\begin{aligned} A &= BD = CE = CDF = BEF & E &= AC = DF = BCD = ABF \\ B &= AD = CF = CDE = AEF & F &= BC = DE = ACD = ABE \\ C &= AE = BF = BDE = ADF & CD &= BE = AF = ABC = ADE = BDF = CEF \\ D &= AB = EF = BCE = ACF \end{aligned}$$

- (f) 2^{6-2} ; 1/4 fraction of
6 factors in 16 runs

Resolution IV

Design Generators

$$E = ABC \quad F = BCD$$

$$\text{Defining relation: } I = ABCE = BCDF = ADEF$$

Aliases

$$\begin{aligned} A &= BCE = DEF & AB &= CE \\ B &= ACE = CDF & AC &= BE \\ C &= ABE = BDF & AD &= EF \\ D &= BCF = AEF & AE &= BC = DF \\ E &= ABC = ADF & AF &= DE \\ F &= BCD = ADE & BD &= CF \\ ABD &= CDE = ACF = BEF & BF &= CD \\ ACD &= BDE = ABF = CEF \end{aligned}$$

2 blocks of 8: $ABD = CDE = ACF = BEF$

- (g) 2^{6-1} ; 1/2 fraction of
6 factors in 32 runs

Resolution VI

Design Generators

$$F = ABCDE$$

$$\text{Defining relation: } I = ABCDEF$$

Aliases

Each main effect is aliased with a single 5-factor interaction.

Each 2-factor interaction is aliased with a single 4-factor interaction.

$$\begin{aligned} ABC &= DEF & ACE &= BDF \\ ABD &= CEF & ACF &= BDE \\ ABE &= CDF & ADE &= BCF \\ ABF &= CDE & ADF &= BCE \\ ACD &= BEF & AEF &= BCD \end{aligned}$$

2 blocks of 16: $ABC = DEF$ 4 blocks of 8: $AB = CDEF$

$$\begin{aligned} ACD &= BEF \\ AEF &= BCD \end{aligned}$$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 7 Factors

- (h)
- 2^{7-4}
- ; 1/16 fraction of
-
- 7 factors in 8 runs

Resolution III

Design Generators

$$D = AB \quad E = AC \quad F = BC \quad G = ABC$$

$$\text{Defining relation: } I = ABD = ACE = BCDE = BCF = ACDF = ABEF = DEF = ABCG \\ = CDG = BEG = ADEG = AFG = BDFG = CEFG = ABCDEFG$$

Aliases

$$\begin{aligned} A &= BD = CE = FG & E &= AC = DF = BG \\ B &= AD = CF = EG & F &= BC = DE = AG \\ C &= AE = BF = DG & G &= CD = BE = AF \\ D &= AB = EF = CG \end{aligned}$$

- (i)
- 2^{7-3}
- ; 1/8 fraction of 7
-
- factors in 16 runs

Resolution IV

Design Generators

$$E = ABC \quad F = BCD \quad G = ACD$$

$$\text{Defining relation: } I = ABCE = BCDF = ADEF = ACDG = BDEG = ABFG = CEFG$$

Aliases

$$\begin{aligned} A &= BCE = DEF = CDG = BFG & AB &= CE = FG & E &= ABC = ADF = BDG = CFG & AF &= DE = BG \\ B &= ACE = CDF = DEG = AFG & AC &= BE = DG & F &= BCD = ADE = ABG = CEG & AG &= CD = BF \\ C &= ABE = BDF = ADG = EFG & AD &= EF = CG & G &= ACD = BDE = ABF = CEF & BD &= CF = EG \\ D &= BCF = AEF = ACG = BEG & AE &= BC = DF \end{aligned}$$

$$ABD = CDE = ACF = BEF = BCG = AEG = DFG$$

$$2 \text{ blocks of 8: } ABD = CDE = ACF = BEF = BCG = AEG = DFG$$

- (j)
- 2^{7-2}
- ; 1/4 fraction of 7
-
- factors in 32 runs

Resolution IV

Design Generators

$$F = ABCD \quad G = ABDE$$

$$\text{Defining relation: } I = ABCDF = ABDEG = CEFG$$

Aliases

$$\begin{aligned} A &= & AB &= CDF = DEG & BC &= ADF & CE &= FG & ACE &= AFG \\ B &= & AC &= BDF & BD &= ACF = AEG & CF &= ABD = EG & ACG &= AEF \\ C &= EFG & AD &= BCF = BEG & BE &= ADG & CG &= EF & BCE &= BFG \\ D &= & AE &= BDG & BF &= ACD & DE &= ABG & BCG &= BEF \\ E &= CFG & AF &= BCD & BG &= ADE & DF &= ABC & CDE &= DFG \\ F &= CEG & AG &= BDE & CD &= ABF & DG &= ABE & CDG &= DEF \\ G &= CEF \end{aligned}$$

$$2 \text{ blocks of 16: } ACE = AFG \quad 4 \text{ blocks of 8: } ACE = AFG$$

$$BCE = BFG$$

$$AB = CDF = DEG$$

- (k)
- 2^{7-1}
- ; 1/2 fraction of
-
- 7 factors in 64 runs

Resolution VII

Design Generators

$$G = ABCDEF$$

$$\text{Defining relation: } I = ABCDEFG$$

Aliases

Each main effect is aliased with a single 6-factor interaction.

Each 2-factor interaction is aliased with a single 5-factor interaction.

Each 3-factor interaction is aliased with a single 4-factor interaction.

$$2 \text{ blocks of 32: } ABC \quad 4 \text{ blocks of 16: } ABC$$

$$CEF$$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 8 Factors

- (l) 2^{8-4} ; 1/16 fraction of
8 factors in 16 runs

Resolution IV

Design Generators

$$E = BCD \quad F = ACD \quad G = ABC \quad H = ABD$$

$$\text{Defining relation: } I = BCDE = ACDF = ABEF = ABCG = ADEG = BDFG = CEFG = ABDH \\ = ACEH = BCFH = DEFH = CDGH = BEGH = AFGH = ABCDEFGH$$

Aliases

$$\begin{aligned} A &= CDF = BEF = BCG = DEG = BDH = CEH = FGH & AB &= EF = CG = DH \\ B &= CDE = AEF = ACG = DFG = ADH = CFH = EGH & AC &= DF = BG = EH \\ C &= BDE = ADF = ABG = EFG = AEH = BFH = DGH & AD &= CF = EG = BH \\ D &= BCE = ACF = AEG = BFG = ABH = EFH = CGH & AE &= BF = DG = CH \\ E &= BCD = ABF = ADG = CFG = ACH = DFH = BGH & AF &= CD = BE = GH \\ F &= ACD = ABE = BDG = CEG = BCH = DEH = AGH & AG &= BC = DE = FH \\ G &= ABC = ADE = BDF = CEF = CDH = BEH = AFH & AH &= BD = CE = FG \\ H &= ABD = ACE = BCF = DEF = CDG = BEG = AFG \end{aligned}$$

$$2 \text{ blocks of } 8: AB = EF = CG = DH$$

- (m) 2^{8-3} ; 1/8 fraction of
8 factors in 32 runs

Resolution IV

Design Generators

$$F = ABC \quad G = ABD \quad H = BCDE$$

$$\text{Defining relation: } I = ABCF = ABDG = CDFG = BCDEH = ADEFH = ACEGH = BEFGH$$

Aliases

$$\begin{aligned} A &= BCF = BDG & AE &= DFH = CGH & DE &= BCH = AFH \\ B &= ACF = ADG & AF &= BC = DEH & DH &= BCE = AEF \\ C &= ABF = DFG & AG &= BD = CEH & EF &= ADH = BGH \\ D &= ABG = CFG & AH &= DEF = CEG & EG &= ACH = BFH \\ E &= & BE &= CDH = FGH & EH &= BCD = ADF = ACG = BFG \\ F &= ABC = CDG & BH &= CDE = EFG & FH &= ADE = BEG \\ G &= ABD = CDF & CD &= FG = BEH & GH &= ACE = BEF \\ H &= & CE &= BDH = AGH & ABE &= CEF = DEG \\ AB &= CF = DG & CG &= DF = AEH & ABH &= CFH = DGH \\ AC &= BF = EGH & CH &= BDE = AEG & ACD &= BDF = BCG = AFG \\ AD &= BG = EFH \end{aligned}$$

$$2 \text{ blocks of } 16: ABE = CEF = DEG \quad 4 \text{ blocks of } 8: ABE = CEF = DEG$$

$$ABH = CFH = DGH$$

$$EH = BCD = ADF = ACG = BFG$$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 8 Factors (Continued)

(n) 2^{8-2} ; 1/4 fraction of 8 factors in 64 runs

Resolution V

Design Generators

$$G = ABCD \quad H = ABEF$$

$$\text{Defining relation: } I = ABCDG = ABEFH = CDEFGH$$

Aliases

$$\begin{array}{llll}
 AB = CDG = EFH & BG = ACD & EF = ABH & ADH = BFG = \\
 AC = BDG & BH = AEF & EG = & AEG = BGH = \\
 AD = BCG & CD = ABG & EH = ABF & AFG = CDE = FGH \\
 AE = BFH & CE = & FG = & AGH = CDF = EGH \\
 AF = BEH & CF = & FH = ABE & BCE = CDH = EFG \\
 AG = BCD & CG = ABD & GH = & BCF = CEF = DGH \\
 AH = BEF & CH = & ACE = & BCH = CEG = DFH \\
 BC = ADG & DE = & ACF = & BDE = CEH = DFG \\
 BD = ACG & DF = & ACH = & BDF = CFG = DEH \\
 BE = AFH & DG = ABC & ADE = & BDH = CFH = DEG \\
 BF = AEH & DH = & ADF = & BEG = CGH = DEF
 \end{array}$$

$$2 \text{ blocks of } 32: CDE = FGH \quad 4 \text{ blocks of } 16: CDE = FGH$$

$$ACF$$

$$BDH$$

Designs with 9 Factors

(o) 2^{9-5} ; 1/32 fraction of 9 factors in 16 runs

Resolution III

Design Generators

$$E = ABC \quad F = BCD \quad G = ACD \quad H = ABD \quad J = ABCD$$

$$\begin{aligned}
 \text{Defining relation: } I &= ABCE = BCDF = ADEF = ACDG = BDEG = ABFG = CEFG = ABDH \\
 &= CDEH = ACFH = BEFH = BCGH = AEGH = DFGH = ABCDEFGH = ABCDJ \\
 &= DEJ = AFJ = BCEFJ = BGJ = ACEGJ = CDFGJ = ABDEFGJ = CHJ \\
 &= ABEHJ = BDFHJ = ACDEFHJ = ADGHJ = BCDEGHJ = ABCFGHJ = EFGHJ
 \end{aligned}$$

Aliases

$$\begin{aligned}
 A &= FJ \\
 B &= GJ \\
 C &= HJ \\
 D &= EJ \\
 E &= DJ \\
 F &= AJ \\
 G &= BJ \\
 H &= CJ \\
 J &= DE = AF = BG = CH
 \end{aligned}$$

$$AB = CE = FG = DH$$

$$AC = BE = DG = FH$$

$$AD = EF = CG = BH$$

$$AE = BC = DF = GH$$

$$AG = CD = BF = EH$$

$$AH = BD = CF = EG$$

$$2 \text{ blocks of } 8: AB = CE = FG = DH$$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 9 Factors (Continued)

(p) 2^{9-4} ; 1/16 fraction of
9 factors in 32 runs

Resolution IV

Design Generators

$$F = BCDE \quad G = ACDE \quad H = ABDE \quad J = ABCE$$

$$\text{Defining relation: } I = BCDEF = ACDEG = ABFG = ABDEH = ACFH = BCGH = DEFGH = ABCEJ \\ = ADFJ = BDGJ = CEFGJ = CDHJ = BEFHJ = AEGHJ = ABCDFGHJ$$

Aliases

$$\begin{array}{lll} A = BFG = CFH = DFJ & AD = CEG = BEH = FJ & BJ = ACE = DG = EFH \\ B = AFG = CGH = DGJ & AE = CDG = BDH = BCJ = GHJ & CD = BEF = AEG = HJ \\ C = AFH = BGH = DHJ & AF = BG = CH = DJ & CE = BDF = ADG = ABJ = FGJ \\ D = AFJ = BGJ = CHJ & AG = CDE = BF = EHJ & CJ = ABE = EFG = DH \\ E = & AH = BDE = CF = EGJ & DE = BCF = ACG = ABH = FGH \\ F = ABG = ACH = ADJ & AJ = BCE = DF = EGH & EF = BCD = DGH = CGJ = BHJ \\ G = ABF = BCH = BDJ & BC = DEF = GH = AEJ & EG = ACD = DFH = CFJ = AHJ \\ H = ACF = BCG = CDJ & BD = CEF = AEH = GJ & EH = ABD = DFG = BFJ = AGJ \\ J = ADF = BDG = CDH & BE = CDF = ADH = ACJ = FHJ & EJ = ABC = CFG = BFH = AGH \\ AB = FG = DEH = CEJ & BH = ADE = CG = EFJ & AEF = BEG = CEH = DEJ \\ AC = DEG = FH = BEJ & & \end{array}$$

$$2 \text{ blocks of } 16: AEF = BEG = CEH = DEJ \quad 4 \text{ blocks of } 8: AEF = BEG = CEH = DEJ$$

$$AB = FG = DEH = CEJ$$

$$CD = BEF = AEG = HJ$$

(q) 2^{9-3} ; 1/8 fraction of 9
factors in 64 runs

Resolution IV

Design Generators

$$G = ABCD \quad H = ACEF \quad J = CDEF$$

$$\text{Defining relation: } I = ABCDG = ACEFH = BDEFGH = CDEFJ = ABEFGJ = ADHJ = BCGHJ$$

Aliases

$$\begin{array}{lll} A = DHJ & AC = BDG = EFH & BF = \\ B = & AD = BCG = HJ & BG = ACD = CHJ \\ C = & AE = CFH & BH = CGJ \\ D = AHJ & AF = CEH & BJ = CGH \\ E = & AG = BCD & CD = ABG = EFJ \\ F = & AH = CEF = DJ & CE = AFH = DFJ \\ G = & AJ = DH & CF = AEH = DEJ \\ H = ADJ & BC = ADG = GHJ & CG = ABD = BHJ \\ J = ADH & BD = ACG & CH = AEF = BGJ \\ AB = CDG & BE = & CJ = DEF = BGH \\ DE = CFJ & GJ = BCH & AFJ = BEG = DFH \\ DF = CEJ & ABE = FGJ & AGH = DGJ \\ DG = ABC & ABF = EGJ & AGJ = BEF = DGH \\ EF = ACH = CDJ & ABH = BDJ & BCE = \\ EG = & ABJ = EFG = BDH & BCF = \\ EH = ACF & ACJ = CDH & BDE = FGH \\ EJ = CDF & ADE = EHJ & BDF = EGH \\ FG = & ADF = FHJ & BEH = DFG \\ FH = ACE & AEG = BFJ & BFH = DEG \\ FJ = CDE & AEJ = BFG = DEH & CEG = \\ GH = BCJ & AFG = BEJ & CFG = \end{array}$$

$$2 \text{ blocks of } 32: CFG \quad 4 \text{ blocks of } 16: CFG =$$

$$AGJ = BEF = DGH$$

$$ADE = EHJ$$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 10 Factors

(r) 2^{10-6} , 1/64 fraction of
10 factors in 16 runs

Resolution III

Design Generators

$$E = ABC \quad F = BCD \quad G = ACD \quad H = ABD \quad J = ABCD \quad K = AB$$

Defining relation: $I = ABCE = BCDF = ADEF = ACDG = BDEG = ABFG = CEFG = ABDH$
 $= CDEH = ACFH = BEFH = BCGH = AEGH = DFGH = ABCDEFGH = ABCDJ$
 $= DEJ = AFJ = BCEFJ = BGJ = ACEGJ = CDFGI = ABDEFGJ = CHJ$
 $= ABEHJ = BDFHJ = ACDEFHJ = ADGHJ = BCDEGHJ = ABCFGHJ = EFGHJ = ABK$
 $= CEK = ACDFK = BDEFK = BCDGK = ADEGK = FGK = ABCEFGK = DHK$
 $= ABCDEHK = BCFHK = AEFHK = ACGHK = BEGHK = ABDFGHK = CDEFGHK = CDJK$
 $= ABDEJK = BFJK = ACEFJK = AGJK = BCEGJK = ABCDFGJK = DEFGJK = ABCHJK$
 $= EHJK = ADFHJK = BCDEFHJK = BDGHJK = ACDEGHJK = CFGHJK = ABEFGHJK$

Aliases

$$\begin{aligned} A &= FJ = BK & J &= DE = AF = BG = CH \\ B &= GJ = AK & K &= AB = CE = FG = DH \\ C &= HJ = EK & AC &= BE = DG = FH \\ D &= EJ = HK & AD &= EF = CG = BH \\ E &= DJ = CK & AE &= BC = DF = GH \\ F &= AJ = GK & AG &= CD = BF = EH = JK \\ G &= BJ = FK & AH &= BD = CF = EG \\ H &= CJ = DK \end{aligned}$$

2 blocks of 8: $AG = CD = BF = EH = JK$

(s) 2^{10-5} , 1/32 fraction of
10 factors in 32 runs

Resolution IV

Design Generators

$$F = ABCD \quad G = ABCE \quad H = ABDE \quad J = ACDE \quad K = BCDE$$

Defining relation: $I = ABCDF = ABCEG = DEFG = ABDEH = CEFH = CDGH = ABFGH = ACDEJ$
 $= BEFJ = BDGJ = ACFGJ = BCHJ = ADFHJ = AEGHJ = BCDEFGHJ = BCDEK$
 $= AEFK = ADGK = BCFGK = ACHK = BDFHK = BEGHK = ACDEFGHK = ABJK$
 $= CDFJK = CEGJK = ABDEFGJK = DEHJK = ABCEFHJK = ABCDGHJK = FGHJK$

Aliases

$$\begin{aligned} A &= EFK = DGK = CHK = BJK & AH &= BDE = BFG = DFJ = EGJ = CK \\ B &= EFJ = DGJ = CHJ = AJK & AJ &= CDE = CFG = DFH = EGH = BK \\ C &= EFH = DGH = BHJ = AHK & AK &= EF = DG = CH = BJ \\ D &= EFG = CGH = BGJ = AGK & BC &= ADF = AEG = HJ = DEK = FGK \\ E &= DFG = CFH = BFJ = AFK & BD &= ACF = AEH = GJ = CEK = FHK \\ F &= DEG = CEH = BEJ = AEK & BE &= ACG = ADH = FJ = CDK = GHK \\ G &= DEF = CDH = BDJ = ADK & BF &= ACD = AGH = EJ = CGK = DHK \\ H &= CEF = CDG = BCJ = ACK & BG &= ACE = AFH = DJ = CFK = EHK \\ J &= BEF = BDG = BCH = ABK & BH &= ADE = AFG = CJ = DFK = EGK \\ K &= AEF = ADG = ACH = ABJ & CD &= ABF = GH = AEJ = BEK = FJK \\ AB &= CDF = CEG = DEH = FGH = JK & CE &= ABG = FH = ADJ = BDK = GJK \\ AC &= BDF = BEG = DEJ = FGJ = HK & CF &= ABD = EH = AGJ = BGK = DJK \\ AD &= BCF = BEH = CEJ = FHJ = GK & CG &= ABE = DH = AFJ = BFK = EJK \\ AE &= BCG = BDH = CDJ = GHJ = FK & DE &= FG = ABH = ACJ = BCK = HJK \\ AF &= BCD = BGH = CGJ = DHJ = EK & DF &= ABC = EG = AHJ = BHK = CJK \\ AG &= BCE = BFH = CFJ = EHJ = DK \end{aligned}$$

2 blocks of 16: $AK = EF = DG = CH = BJ$

4 blocks of 8: $AK = EF = DG = CH = BJ$

$AJ = CDE = CFG = DFH = EGH = BK$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 10 Factors (Continued)

(t) 2^{10-4} ; 1/16 fraction of
10 factors in 64 runs

Resolution IV

Design Generators

$$G = BCDF \quad H = ACDF \quad J = ABDE \quad K = ABCE$$

$$\text{Defining relation: } I = BCDFG = ACDFH = ABGH = ABDEJ = ACEFGJ = BCEFHJ = DEGHJ = ABCEK \\ = ADEFGK = BDEFHK = CEGHK = CDJK = BFGJK = AFHJK = ABCDGHJK$$

Aliases

$A = BGH$	$AD = CFH = BEJ$	$BK = ACE = FGJ$
$B = AGH$	$AE = BDJ = BCK$	$CD = BFG = AFH = JK$
$C = DJK$	$AF = CDH = HJK$	$CE = ABK = GHK$
$D = CJK$	$AG = BH$	$CF = BDG = ADH$
$E =$	$AH = CDF = BG = FJK$	$CG = BDF = EHK$
$F =$	$AJ = BDE = FHK$	$CH = ADF = EGK$
$G = ABH$	$AK = BCE = FHJ$	$CJ = DK$
$H = ABG$	$BC = DFG = AEK$	$CK = ABE = EGH = DJ$
$J = CDK$	$BD = CFG = AEJ$	$DE = ABJ = GHJ$
$K = CDJ$	$BE = ADJ = ACK$	$DF = BCG = ACH$
$AB = GH = DEJ = CEK$	$BF = CDG = GJK$	$DG = BCF = EHJ$
$AC = DFH = BEK$	$BJ = ADE = FGK$	$DH = ACF = EGJ$
$EF =$	$GJ = DEH = BFK$	$AEG = BEH = CFJ = DFK$
$EG = DHJ = CHK$	$GK = CEH = BFJ$	$AEH = BEG$
$EH = DGJ = CGK$	$HJ = DEG = AFK$	$AFG = BFH = CEJ = DEK$
$EJ = ABD = DGH$	$HK = CEG = AFJ$	$AGJ = CEF = BHJ$
$EK = ABC = CGH$	$ABF = FGH$	$AGK = DEF = BHK$
$FG = BCD = BJK$	$ACG = BCH = EFJ$	$BCJ = EFH = BDK$
$FH = ACD = AJK$	$ACJ = EFG = ADK$	$BEF = CHJ = DHK$
$FJ = BGK = AHK$	$ADG = BDH = EFK$	$CDE = EJK$
$FK = BGJ = AHJ$	$AEF = CGJ = DGK$	$CFK = DFJ$

2 blocks of 32: $AGJ = CEF = BHJ$ 4 blocks of 16: $AGJ = CEF = BHJ$ $AGK = DEF = BHK$ $CD = BFG = AFH = JK$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 11 Factors

(u) 2^{11-7} ; 1/128 fraction of
11 factors in 16 runs

Resolution III

Design Generators

$$E = ABC \quad F = BCD \quad G = ACD \quad H = ABD \quad J = ABCD \quad K = AB \quad L = AC$$

Defining relation: $I = ABCE = BCDF = ADEF = ACDG = BDEG = ABFG = CEF = ABDH$
 $= CDEH = ACFH = BEFH = BCGH = AEGH = DFGH = ABCDEFGH = ABCDJ$
 $= DEJ = AFJ = BCEFJ = BGJ = ACEGJ = CDFGJ = ABDEFGJ = CHJ$
 $= ABEHJ = BDFHJ = ACDEFHJ = ADGHJ = BCDEGHJ = ABCFGHJ = EFGHJ = ABK$
 $= CEK = ACDFK = BDEFK = BCDGK = ADEGK = FGK = ABCEFGK = DHK$
 $= ABCDEHK = BCFHK = AEFHK = ACGHK = BEGHK = ABDFGHK = CDEFGHK = CDJK$
 $= ABDEJK = BFJK = ACEFJK = AGJK = BCEGJK = ABCDFGJK = DEFGJK = ABCHJK$
 $= EHJK = ADFHJK = BCDEFHJK = BDGHJK = ACDEGHJK = CFGHJK = ABEFGHJK = ACL$
 $= BEL = ABDFL = CDEFL = DGL = ABCDEGL = BCFGL = AEFGL = BCDHL$
 $= ADEHL = FHL = ABCEFHL = ABGHL = CEGHL = ACDFGHL = BDEFGHL = BDJL$
 $= ACDEJL = CFJL = ABEFJL = ABCGJL = EGJL = ADFGJL = BCDEFGJL = AHJL$
 $= BCEHJL = ABCDFHJL = DEFHJL = CDGHJL = ABDEGHJL = BFGHJL = ACEFGHJL = BCKL$
 $= AEKL = DFKL = ABCDEFKL = ABDGKL = CDEGKL = ACFGKL = BEFGKL = ACDHKL$
 $= BDEHKL = ABFHKL = CEFHKL = GHKL = ABCEGHKL = BCDFGHKL = ADEFGHKL = ADJKL$
 $= BCDEJKL = ABCFJKL = EFJKL = CGJKL = ABEGJKL = BDFGJKL = ACDEFGJKL = BHJKL$
 $= ACEHJKL = CDFHJKL = ABDEFHJKL = ABCDGHJKL = DEGHJKL = AFGHJKL = BCEFGHJKL$

Aliases

$$\begin{array}{ll} A = FJ = BK = CL & J = DE = AF = BG = CH \\ B = GJ = AK = EL & K = AB = CE = FG = DH \\ C = HJ = EK = AL & L = AC = BE = DG = FH \\ D = EJ = HK = GL & AD = EF = CG = BH \\ E = DJ = CK = BL & AE = BC = DF = GH = KL \\ F = AJ = GK = HL & AG = CD = BF = EH = JK \\ G = BJ = FK = DL & AH = BD = CF = EG = JL \\ H = CJ = DK = FL & \end{array}$$

2 blocks of 8: $AE = BC = DF = GH = KL$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 11 Factors (Continued)

(v) 2^{11-6} ; 1/64 fraction of
11 factors in 32 runs

Resolution IV

Design Generators

$$F = ABC \quad G = BCD \quad H = CDE \quad J = ACD \quad K = ADE \quad L = BDE$$

Defining relation:

$$\begin{aligned} I &= ABCF = BCDG = ADFG = CDEH = ABDEFH = BEGH = ACEFGH = ACDJ = BDFJ = ABGJ = CFGJ \\ &= AEHJ = BCEFHJ = ABCDEGHJ = DEFGHJ = ADEK = BCDEFK = ABCEGK = EFGK = ACHK = BFHK \\ &= ABDGHK = CDFGHK = CEJK = ABEFJK = BDEGJK = ACDEFGJK = DHJK = ABCDFHJK = BCGHJK \\ &= AFGHJK = BDEL = ACDEFL = CEGJ = ABEFGL = BCHL = AFHL = DGHL = ABCDFGHL \\ &= ABCEJL = EFJL = ADEGJL = BCDEFGJL = ABDHJL = CDFHJL = ACGHJL = BFGHJL = ABKL \\ &= CFKL = ACDGKL = BDFGKL = ABCDEHKL = DEFHKL = AEGHKL = BCEFGHKL = BCDJKL \\ &= ADFJKL = GJKL = ABCFGJKL = BEHJKL = ACEFHJKL = CDEGHJKL = ABDEFGHJKL \end{aligned}$$

Aliases

$$\begin{aligned} A &= BCF = DFG = CDJ = BGJ = EHJ = DEK = CHK = FHL = BKL \\ B &= ACF = CDG = EGH = DFJ = AGJ = FHK = DEL = CHL = AKL \\ C &= ABF = BDG = DEH = ADJ = FGJ = AHK = EJK = EGL = BHL = FKL \\ D &= BCG = AFG = CEH = ACJ = BFJ = AEK = HJK = BEL = GHL \\ E &= CDH = BGH = AHJ = ADK = FGK = CJK = BDL = CGL = FJL \\ F &= ABC = ADG = BDJ = CGJ = EGK = BHK = AHL = EKL = CKL \\ G &= BCD = ADF = BEH = ABJ = CFJ = EFK = CEL = DHL = JKL \\ H &= CDE = BEG = AEJ = ACK = BFK = DJK = BCL = AFL = DGL \\ J &= ACD = BDF = ABG = CFG = AEH = CEK = DHK = EFL = GKL \\ K &= ADE = EFG = ACH = BFH = CEJ = DHJ = ABL = CFL = GJL \\ L &= BDE = CEG = BCH = AFH = DGH = EFJ = ABK = CFK = GJK \\ AB &= CF = GJ = KL \quad AE = HJ = DK \quad AH = EJ = CK = FL \quad AL = FH = BK \quad BH = EG = CL = FK \\ AC &= BF = DJ = HK \quad AF = BC = DG = HL \quad AJ = CD = BG = EH \quad BD = CG = FJ = EL \quad CE = DH = JK = GL \\ AD &= FG = CJ = EK \quad AG = DF = BJ \quad AK = DE = CH = BL \quad BE = GH = DL \quad EF = GK = JL \\ ABD &= CDF = ACG = BFG = EFH = BCJ = AFJ = DGJ = BEK = GHK = AEL = HJL = DKL \\ ABE &= CEF = DFH = AGH = EGJ = BHJ = BDK = CGK = FJK = ADL = FGL = CJL = EKL \\ ABH &= DEF = AEG = CFH = BEJ = GHJ = BCK = AFK = DGK = ACL = BFL = DJL = HKL \\ ACE &= BEF = ADH = FGH = DEJ = CHJ = CDK = BGK = EHK = AJK = DFL = AGL = BJL \\ AEF &= BCE = DEG = BDH = CGH = FHJ = DFK = AGK = BJK = CDL = BGL = EHL = AJL \\ 2 \text{ blocks of } 16: & AB = CF = GJ = KL \quad 4 \text{ blocks of } 8: \quad AB = CF = GJ = KL \\ & AD = FG = CJ = EK \\ & BD = CG = FJ = EL \end{aligned}$$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (Continued)

Designs with 12 Factors

(w) 2^{12-8} ; 1/256 fraction
of 12 factors in 16 runs

Resolution IIIDesign Generators

$$E = ABC \quad F = ABD \quad G = ACD \quad H = BCD \\ J = ABCD \quad K = AB \quad L = AC \quad M = AD$$

Aliases

$$\begin{aligned} A &= HJ = BK = CL = DM \\ B &= GJ = AK = EL = FM \\ C &= FJ = EK = AL = GM \\ D &= EJ = FK = GL = AM \\ E &= DJ = CK = BL = HM \\ F &= CJ = DK = HL = BM \\ G &= BJ = HK = DL = CM \\ H &= AJ = GK = FL = EM \\ J &= DE = CF = BG = AH \\ K &= AB = CE = DF = GH \\ L &= AC = BE = DG = FH \\ M &= AD = BF = CG = EH \\ AE &= BC = FG = DH = KL = JM \\ AF &= BD = EG = CH = JL = KM \\ AG &= EF = CD = BH = JK = LM \end{aligned}$$

2 blocks of 8: $AE = BC = FG = DH = KL = JM$

Designs with 13 Factors

(x) 2^{13-9} ; 1/512 fraction of
13 factors in 16 runs

Resolution IIIDesign Generators

$$E = ABC \quad F = ABD \quad G = ACD \quad H = BCD \\ J = ABCD \quad K = AB \quad L = AC \quad M = AD \quad N = BC$$

Aliases

$$\begin{aligned} A &= HJ = BK = CL = DM = EN \\ B &= GJ = AK = EL = FM = CN \\ C &= FJ = EK = AL = GM = BN \\ D &= EJ = FK = GL = AM = HN \\ E &= DJ = CK = BL = HM = AN \\ F &= CJ = DK = HL = BM = GN \\ G &= BJ = HK = DL = CM = FN \\ H &= AJ = GK = FL = EM = DN \\ J &= DE = CF = BG = AH = MN \\ K &= AB = CE = DF = GH = LN \\ L &= AC = BE = DG = FH = KN \\ M &= AD = BF = CG = EH = JN \\ N &= BC = AE = FG = DH = KL = JM \\ AF &= BD = EG = CH = JL = KM \\ AG &= EF = CD = BH = JK = LM \end{aligned}$$

2 blocks of 8: $AF = BD = EG = CH = JL = KM$

X Alias Relationships for 2^{k-p} Fractional Factorial Designs with $k \leq 15$ and $n \leq 64$ (*Continued*)

Designs with 14 Factors

(y) 2^{14-10} ; 1/1024 fraction of
14 factors in 16 runs

Resolution III

Design Generators

$$E = ABC \quad F = ABD \quad G = ACD \quad H = BCD \quad J = ABCD \\ K = AB \quad L = AC \quad M = AD \quad N = BC \quad O = BD$$

Aliases

$$\begin{aligned} A &= HJ = BK = CL = DM = EN = FO \\ B &= GJ = AK = EL = FM = CN = DO \\ C &= FJ = EK = AL = GM = BN = HO \\ D &= EJ = FK = GL = AM = HN = BO \\ E &= DJ = CK = BL = HM = AN = GO \\ F &= CJ = DK = HL = BM = GN = AO \\ G &= BJ = HK = DL = CM = FN = EO \\ H &= AJ = GK = FL = EM = DN = CO \\ J &= DE = CF = BG = AH = MN = LO \\ K &= AB = CE = DF = GH = LN = MO \\ L &= AC = BE = DG = FH = KN = JO \\ M &= AD = BF = CG = EH = JN = KO \\ N &= BC = AE = FG = DH = KL = JM \\ O &= BD = AF = EG = CH = JL = KM \\ AG &= EF = CD = BH = JK = LM = NO \end{aligned}$$

2 blocks of 8: $AG = EF = CD = BH = JK = LM = NO$

Designs with 15 Factors

(z) 2^{15-11} ; 1/2048 fraction of
15 factors in 16 runs

Resolution III

Design Generators

$$E = ABC \quad F = ABD \quad G = ACD \quad H = BCD \quad J = ABCD \\ K = AB \quad L = AC \quad M = AD \quad N = BC \quad O = BD \quad P = CD$$

Aliases

$$\begin{aligned} A &= HJ = BK = CL = DM = EN = FO = GP \\ B &= GJ = AK = EL = FM = CN = DO = HP \\ C &= FJ = EK = AL = GM = BN = HO = DP \\ D &= EJ = FK = GL = AM = HN = BO = CP \\ E &= DJ = CK = BL = HM = AN = GO = FP \\ F &= CJ = DK = HL = BM = GN = AO = EP \\ G &= BJ = HK = DL = CM = FN = EO = AP \\ H &= AJ = GK = FL = EM = DN = CO = BP \\ J &= DE = CF = BG = AH = MN = LO = KP \\ K &= AB = CE = DF = GH = LN = MO = JP \\ L &= AC = BE = DG = FH = KN = JO = MP \\ M &= AD = BF = CG = EH = JN = KO = LP \\ N &= BC = AE = FG = DH = KL = JM = OP \\ O &= BD = AF = EG = CH = JL = KM = NP \\ P &= CD = EF = AG = BH = JK = LM = NO \end{aligned}$$
